Engineering Challenges

Section Européenne – Sciences de l'Ingénieur

EC01 to EC30

Contents

EC01 – Renewable Energy

Can solar power meet tomorrow's energy needs?

Learning Objectives

- Explain how a photovoltaic cell and a grid-connected PV system operate.
- Interpret capacity data and calculate annual additions.
- Discuss intermittency, storage, grid integration and recycling.

Engineering News

Global renewable capacity grew by a record 692 GW in 2025. Solar power alone accounted for about 510 GW of additions, nearly three-quarters of the renewable total. The engineering challenge is now not only to manufacture more panels, but also to connect variable generation to stronger and more flexible grids.

Main Article

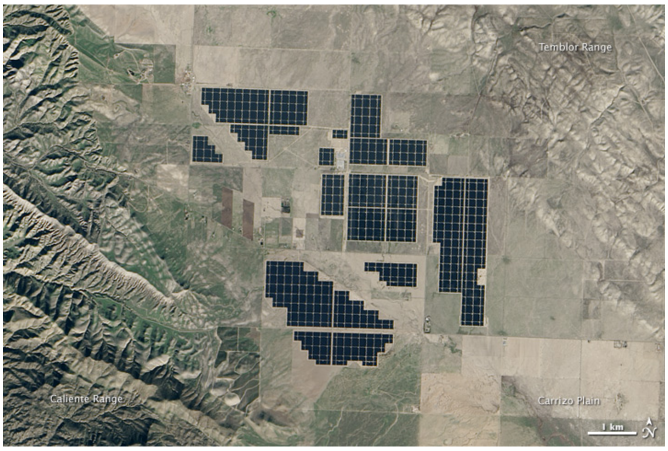
A photovoltaic (PV) cell converts light directly into electricity. Most commercial cells are manufactured from silicon, a semiconductor. When photons provide enough energy, electrons can move through the material. The internal electric field of the p–n junction separates the electrical charges and creates a direct current.

A PV module connects many cells together. Modules are combined into strings and arrays to reach the required voltage and power. However, a module rarely operates at a fixed point. Its voltage-current characteristic changes with solar irradiance and temperature. A maximum power point tracker continuously adjusts the operating point so that the array delivers as much power as possible.

The generated electricity is direct current, whereas buildings and public grids use alternating current. An inverter therefore converts DC into AC, synchronises the voltage and frequency with the grid, and provides protection and monitoring functions. A battery may store surplus electricity, but it increases cost, material use and conversion losses.

Solar power is variable: output changes with clouds, season and time of day. Engineers integrate large amounts of PV using forecasting, flexible demand, interconnections, storage and controllable generation. The objective is not to make every installation independent, but to balance the complete electricity system.

The environmental impact of a PV system must be assessed over its entire life cycle. Manufacturing requires energy and materials, while end-of-life modules contain glass, aluminium, copper, polymers and semiconductor materials. Designing modules that are efficient, durable, repairable and recyclable is therefore a major engineering goal.



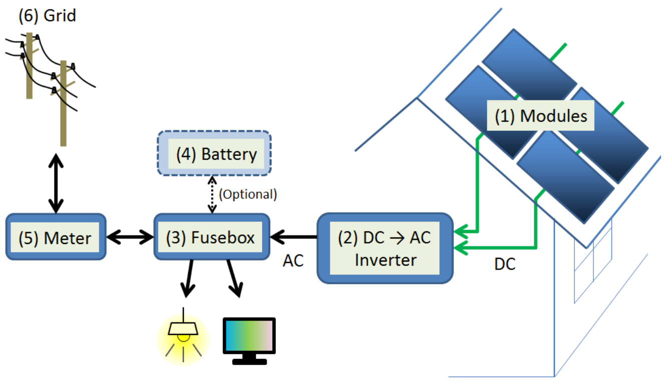
Topaz Solar Farm, California. NASA image, public domain.

Engineering Focus

Designing a photovoltaic installation
Engineers must optimise several parameters simultaneously:

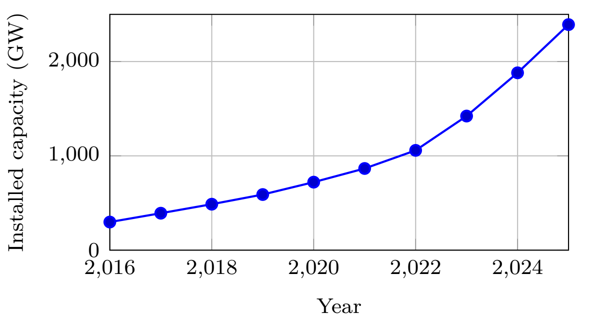
- panel orientation and tilt;
- inverter sizing and MPPT architecture;
- cable cross-sections and electrical losses;
- thermal behaviour and ventilation;
- protection and isolation devices;
- monitoring and maintenance strategy;
- expected lifetime and recyclability.

A good design minimises the levelised cost of electricity while ensuring safety, reliability and high annual energy yield.



Simplified residential PV system. S-kei, CC0.

Global solar PV capacity



Source: IRENA, Renewable Capacity Statistics 2026.

Did You Know?

The solar energy reaching the Earth in roughly one hour is comparable to humanity's annual energy use. The main engineering challenge is therefore to capture, convert, store and distribute only a small fraction of this resource efficiently.

Key Figures

Commercial module efficiency	20–24%
Typical module lifetime	25–30 years
Typical inverter efficiency	97–99%
Annual module degradation	0.3–0.5%/year
Global PV capacity in 2025	≈ 2.4 TW

Useful Vocabulary

solar cell	cellule photovoltaïque
irradiance	irradiance
inverter	onduleur
grid	réseau électrique
storage	stockage
yield	production énergétique
lifetime	durée de vie
recycling	recyclage

Questions

1. Explain the role of the p–n junction.
2. Why does a PV installation need an inverter?
3. What is the purpose of MPPT control?
4. Use the graph to estimate the increase between 2020 and 2025.
5. Give three technical solutions for integrating variable solar power.
6. Why must environmental assessment include manufacturing and end-of-life?

Engineering Challenge

Your school consumes 410 MWh of electricity each year and has 850 m^2 of usable roof area. Prepare a preliminary PV study:

1. estimate the peak power that could be installed;
2. estimate annual energy production;
3. discuss the likely self-consumption ratio;
4. decide whether battery storage is justified;
5. propose useful monitoring indicators.

Present your conclusions as a short engineering report.

Application Exercise

A 420 W PV module receives an average irradiance of 850 W/m^2 over an active area of 2.0 m^2. Calculate its instantaneous efficiency. Then estimate the electrical energy produced by 120 identical modules during 4.5 equivalent full-sun hours. State two reasons why the real daily yield may be lower.

Speaking Task

Working in pairs, prepare a five-minute presentation answering the question: “Can Europe rely mainly on solar electricity by 2050?” Support your arguments with technical evidence, environmental impacts, economic considerations and examples from different European countries.

EC02 – Hydrogen

Can hydrogen decarbonise heavy transport?

Learning Objectives

- Distinguish grey, blue and green hydrogen.
- Explain the operating principles of electrolysers and fuel cells.
- Compare hydrogen and battery-electric energy chains.

Engineering News

Low-emission hydrogen projects are being developed for fertiliser production, steel, shipping and heavy road transport. Their success depends not only on electrolyser capacity, but also on electricity supply, storage, transport and final demand.

Main Article

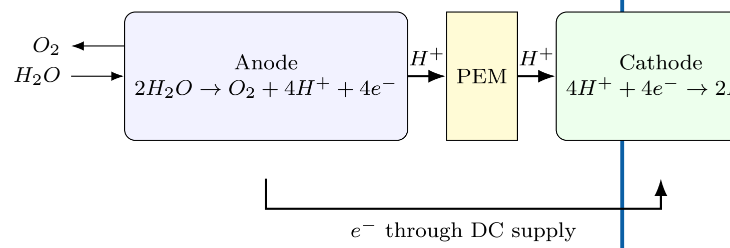
Hydrogen is an **energy carrier**, not a primary source of energy. It must be manufactured from another resource before it can be stored and used. Most hydrogen is still produced from natural gas by steam methane reforming. If the resulting carbon dioxide is released, the product is called grey hydrogen. Capturing part of the CO_2 leads to so-called blue hydrogen. Green hydrogen is produced by electrolysis powered by low-carbon renewable electricity.

In a proton-exchange-membrane (PEM) electrolyser, water enters the anode side. An electric current separates the water molecules into oxygen, protons and electrons. The membrane allows protons to cross while forcing electrons through an external circuit. Hydrogen is formed at the cathode.

A PEM fuel cell performs the reverse overall reaction. Hydrogen is supplied to the anode and oxygen from air reaches the cathode. The electrochemical reaction produces electricity, water and heat without combustion. Several cells are assembled in a stack to obtain the required voltage and power.

Hydrogen has a high specific energy by mass and enables rapid refuelling. However, its low volumetric density requires compression, liquefaction or conversion into another carrier. Each conversion consumes energy. Consequently, a battery-electric vehicle is usually more efficient from electricity source to wheels, while hydrogen can remain attractive when long range, high payload and short refuelling stops are essential.

PEM electrolyser



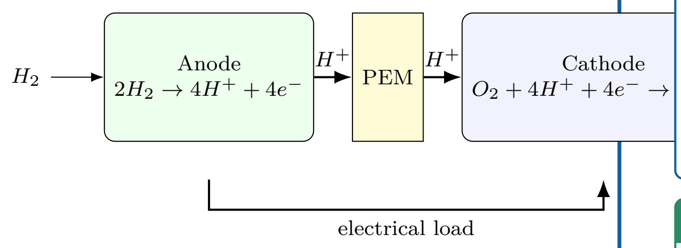
Engineering Focus

The complete hydrogen value chain

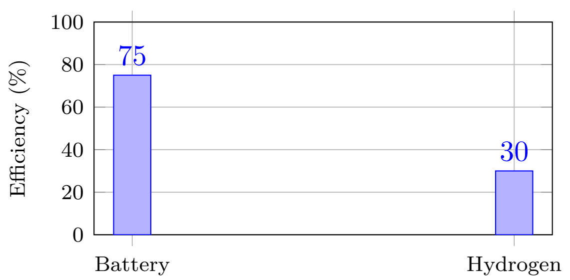
1. Generate low-carbon electricity.
2. Produce hydrogen by electrolysis.
3. Purify and compress the gas.
4. Store and distribute it safely.
5. Convert it into useful energy in a fuel cell or industrial process.

Engineers optimise efficiency, safety, materials, cost and infrastructure at every stage. Hydrogen leaks must also be controlled because the molecule is very small and has a wide flammability range in air.

PEM fuel cell



Electricity-to-wheel efficiency



Illustrative whole-chain values; actual performance depends on equipment and operating conditions.

Did You Know?

Hydrogen contains about 120 MJ of energy per kilogram on a lower-heating-value basis, but storing one kilogram requires a much larger tank than storing an equivalent amount of liquid fuel.

Key Figures

Hydrogen LHV	≈ 120 MJ/kg
Electrolyser efficiency	60–75%
Fuel-cell electrical efficiency	40–60%
Typical vehicle storage pressure	350 or 700 bar

Useful Vocabulary

electrolyser	électrolyseur
fuel cell	pile à combustible
stack	empilement de cellules
compressed gas	gaz comprimé
refuelling	ravitaillement
leak detection	détection de fuite

Questions

1. Why is hydrogen described as an energy carrier?
2. Compare grey, blue and green hydrogen.
3. Explain the role of the PEM in an electrolyser.
4. What products leave a fuel cell?
5. Why is the battery chain usually more efficient?
6. In which transport applications can hydrogen remain attractive?

Engineering Challenge

A logistics company operates 120 long-haul trucks. Compare battery-electric and hydrogen fuel-cell solutions using efficiency, range, payload, refuelling time, infrastructure and life-cycle emissions. Present a justified engineering recommendation.

Application Exercise

An electrolyser consumes 52 kWh of electricity to produce 1 kg of hydrogen. A fuel-cell system later converts the hydrogen into 18 kWh of useful electrical energy. Calculate the power-to-power efficiency of this chain. Compare it with an 88% battery storage system and explain why hydrogen may still be selected for some applications.

Speaking Task

Debate: "Hydrogen will replace batteries for heavy transport." Each team must use at least three technical arguments and respond to one opposing argument.

Sources and further reading

IEA — Global Hydrogen Review: online resource

;

U.S. Department of Energy — Hydrogen and Fuel Cell Technologies Office: online resource

;

European Commission — Hydrogen: online resource

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EC03 – Wind Energy

How do engineers harvest the wind?

Learning Objectives

- Explain how aerodynamic lift produces rotor torque.
- Identify the main subsystems of a modern wind turbine.
- Interpret a wind-turbine power curve.
- Compare onshore, fixed-bottom offshore and floating offshore solutions.

Engineering News

Modern offshore wind turbines now reach ratings above 15 MW. Floating foundations make it possible to exploit deep-water sites where winds are stronger and more regular, but they also introduce new challenges in mooring, maintenance and grid connection.

Key Figures

Rotor diameter	120–250 m
Rated power	3–18 MW
Cut-in speed	3–4 m/s
Rated speed	11–15 m/s
Cut-out speed	20–25 m/s
Capacity factor	30–60%

Main Article

A wind turbine converts the kinetic energy of moving air into electrical energy. The blades are shaped like aerofoils. As air flows around a blade, a pressure difference creates lift. The tangential component of this force produces a torque on the rotor.

The available power in the wind is

$$P_{\text{wind}} = \frac{1}{2}\rho A v^3,$$

where ρ is air density, A the swept area and v wind speed. The cubic term explains why site selection is critical: a small increase in average wind speed can produce a large increase in annual energy output.

The rotor drives either a gearbox and a high-speed generator or a direct-drive generator. Power electronics convert the variable-frequency output into electricity compatible with the grid. Pitch control adjusts the blade angle, while yaw control rotates the nacelle so that the rotor faces the wind.

Engineering Focus

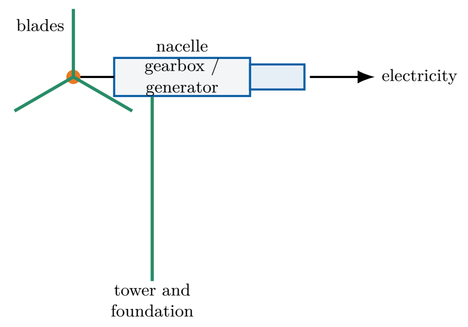
Main subsystems

1. Rotor, blades and hub;
2. Main shaft and bearings;
3. Gearbox or direct-drive generator;
4. Converter and transformer;
5. Pitch and yaw actuators;
6. Tower, foundations and monitoring system.

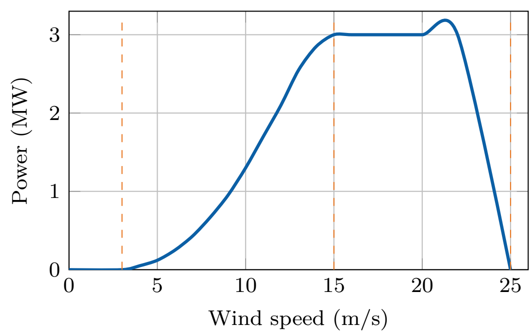
Did You Know?

No turbine can capture all the power in the wind. The theoretical maximum is the **Betz limit**: 59.3% of the kinetic power crossing the rotor area.

Wind-turbine architecture



Typical power curve



Cut-in, rated and cut-out regions determine how the turbine operates safely.

Useful Vocabulary

blade	pale
hub	moyeu
gearbox	multiplicateur
yaw	orientation de la nacelle
pitch	calage des pales
swept area	surface balayée
capacity factor	facteur de charge

Questions

1. Why does wind power depend on the cube of wind speed?
2. What is the purpose of pitch control?
3. What is the purpose of yaw control?
4. Interpret the three regions of the power curve.
5. Compare onshore and offshore wind farms.

Engineering Challenge

A coastal region plans a 500 MW wind farm. Compare an onshore solution, a fixed-bottom offshore solution and a floating offshore solution. Consider energy yield, foundations, maintenance, grid connection, environmental impact and public acceptance. Present a justified recommendation.

Application Exercise

A wind turbine has a rotor diameter of 100 m and operates in air of density 1.225 $\mathrm{kg\,m^{-3}}$ at a wind speed of 10 m/s. Using $P = \frac{1}{2}\rho A v^3 C_p$ with $C_p = 0.44$, estimate the aerodynamic power captured by the rotor. Compare the coefficient with the Betz limit and identify two additional conversion losses before electricity reaches the grid.

Speaking Task

Debate the statement: “Large wind farms should be built mainly offshore.” Use technical, economic and environmental arguments.

Sources

Global Wind Energy Council: online resource

IEA Wind: online resource

EC04 – Nuclear Energy

Can nuclear power support a low-carbon future?

Learning Objectives

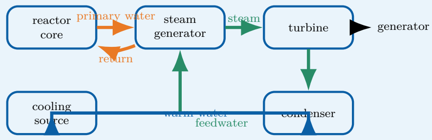
- Explain nuclear fission and the controlled chain reaction.
- Identify the energy and fluid flows in a pressurised water reactor (PWR).
- Interpret lifecycle-emission data and discuss the role of nuclear power.

Engineering News

Several countries are studying new reactor programmes, lifetime extensions and Small Modular Reactors (SMRs). The central engineering questions concern construction time, financing, safety demonstration, fuel supply, waste management and integration with an electricity system containing more variable renewable generation.

Engineering Focus

Energy conversion in a PWR



The primary circuit transports nuclear heat, the secondary circuit produces steam and the cooling circuit rejects unused heat. The reactor is therefore both a nuclear system and a large thermal power plant.

Did You Know?

Stopping fission does not immediately stop heat production. Residual heat falls rapidly, but cooling remains essential for hours, days and longer after shutdown.

Main Article

A nuclear power station converts the energy released by **nuclear fission** into electricity. When a uranium-235 nucleus absorbs a neutron, it can split into two lighter nuclei, release energy and emit additional neutrons. These neutrons may trigger further fissions. In a reactor, the chain reaction is kept stable rather than allowed to grow uncontrolled.

In a pressurised water reactor, fuel assemblies are placed inside the reactor vessel. Water in the primary circuit removes heat from the core. It is maintained at high pressure so that it remains liquid at a temperature of roughly 300 °C. In the steam generator, this primary water transfers heat to a separate secondary circuit. The secondary water boils and produces steam, which drives a turbine connected to an electrical generator.

After the turbine, a condenser transforms the steam back into liquid water. A third cooling circuit carries waste heat towards a river, the sea or a cooling tower. The three circuits have different functions and are kept physically separate.

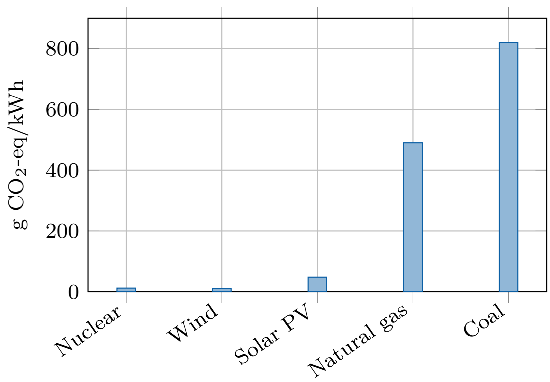
Reactor power is controlled by adjusting neutron absorption. Control rods contain materials that absorb neutrons, while soluble boron may also be used in the primary water. Even after the chain reaction stops, radioactive decay continues to release **residual heat**; cooling must therefore remain available.

Safety relies on several independent barriers and systems: the fuel matrix, the metal cladding, the primary circuit boundary and the containment building. Engineers also analyse extreme events, redundancy, component ageing, human factors, cybersecurity and long-term waste management.

Key Figures

Typical French PWR output	900–1450 MW
Primary-water temperature	about 300 °C
Primary pressure	about 155 bar
Typical capacity factor	70–90%
Design operating life	several decades

Indicative lifecycle emissions



Indicative median lifecycle values based on IPCC assessments. Values vary with technology, location and methodology.

Fission and fusion

Fission splits heavy nuclei and is used in present commercial reactors. **Fusion** combines light nuclei and powers the Sun. Fusion research aims to produce controlled net energy, but it is not yet a commercial electricity source.

Useful Vocabulary

chain reaction	réaction en chaîne
reactor vessel	cuve du réacteur
fuel assembly	assemblage combustible
control rod	barre de contrôle
steam generator	générateur de vapeur
containment	enceinte de confinement
residual heat	puissance résiduelle
spent fuel	combustible usé

Questions

1. What happens during nuclear fission?
2. Why must the primary water remain under high pressure?
3. Explain why the PWR contains several separate water circuits.
4. What is the role of control rods?
5. Why is cooling required after reactor shutdown?
6. Use the graph to compare nuclear power with fossil generation.
7. Give two advantages and two limitations of SMRs.

Engineering Challenge

A region must replace 8 GW of fossil-fuelled generation while reducing lifecycle CO_2 emissions and maintaining electricity supply during winter evenings. Propose a balanced mix using nuclear, wind, solar, storage, interconnections and demand response. Justify:

1. installed capacity;
2. expected availability;
3. flexibility and reserve requirements;
4. environmental and social constraints;
5. construction sequence and major risks.

Present the result as a two-page engineering recommendation.

Application Exercise

A 1.3 GW nuclear unit operates with a capacity factor of 88%. Estimate its annual electrical production in TWh. Then calculate the equivalent annual production of 3 MW wind turbines operating at a 32% capacity factor. Discuss why installed capacity alone is not sufficient for comparing generation technologies.

Speaking Task

Debate: "Europe should build new nuclear power stations." One team supports the proposal and one team opposes it. Use evidence about safety, climate, cost, waste, construction time and security of supply.

Sources and further reading

IAEA - Nuclear Power Reactors: online resource

OECD Nuclear Energy Agency: online resource

IPCC - Energy Systems: online resource

EC05 – Energy Storage

Why storing energy is key to the energy transition

Learning Objectives

- Explain why electrical grids need energy storage.
- Compare storage technologies using power, energy, duration and efficiency.
- Select a suitable storage mix for a given engineering problem.

Engineering News

As variable solar and wind generation expands, grid operators are deploying batteries, pumped-storage plants and other forms of flexibility. The key challenge is not to find one universal storage device, but to combine technologies operating over seconds, hours, days and seasons.

Main Article

Electricity networks must keep production and consumption balanced at every instant. A sudden mismatch changes grid frequency and can damage equipment or trigger disconnections. Storage systems absorb energy when production exceeds demand and return it when the system needs additional power.

Engineers distinguish **power** from **energy**. Power, measured in watts, describes how quickly a storage system can charge or discharge. Energy, measured in watt-hours or joules, describes how long it can maintain that power. A flywheel may deliver high power for a few seconds, whereas a reservoir or hydrogen cavern can store energy for much longer periods.

Lithium-ion batteries provide rapid response and high round-trip efficiency. They are well suited to frequency control, electric vehicles and daily solar shifting. Their limitations include cost, ageing, fire protection and the supply of raw materials. Pumped-storage hydroelectricity stores much larger amounts of energy: water is pumped to an upper reservoir and later released through a reversible turbine. Its lifetime is long, but suitable terrain and major civil works are required.

Compressed air, thermal storage and hydrogen address longer durations. Hydrogen is produced by electrolysis, stored, and later used in a fuel cell or turbine. Because each conversion loses energy, the round-trip efficiency is lower than that of a battery. Nevertheless, hydrogen may become useful when storage duration matters more than efficiency, particularly for seasonal balancing or industrial fuel supply.

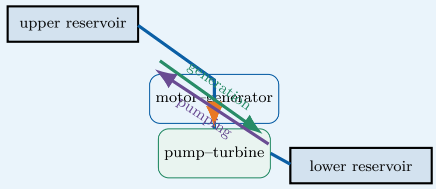
No storage option is optimal for every task. A resilient low-carbon grid therefore combines storage with interconnections, demand response, controllable generation and accurate forecasting.

Did You Know?

A storage system can have excellent efficiency but still be unsuitable if its discharge duration, response time, lifetime or location does not match the service required by the grid.

Engineering Focus

Pumped-storage hydroelectricity

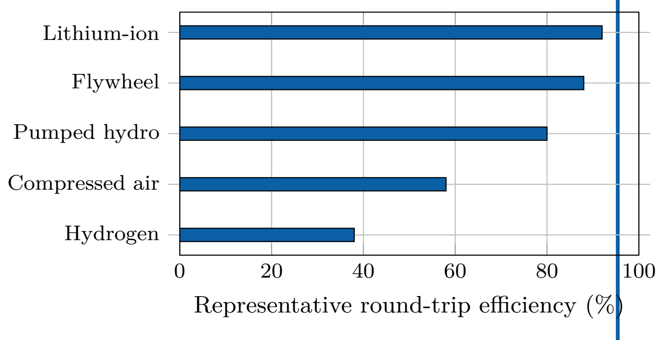


During low demand, electrical energy drives the pump and raises water. During high demand, the water flows downhill and drives the turbine. The stored gravitational energy is $E = mgh$.

Key Figures

Lithium-ion round-trip efficiency	85–95%
Pumped hydro	70–85%
Hydrogen power-to-power	30–45%
Typical battery duration	1–8 h
Seasonal storage	weeks–months

Storage technologies compared



Representative values only: actual performance depends on design and operating conditions.

Useful Vocabulary

round-trip efficiency	rendement aller-retour
charge / discharge	charge / décharge
storage duration	durée de stockage
reservoir	réservoir
flywheel	volant d'inertie
demand response	effacement de consommation
grid frequency	fréquence du réseau

Questions

1. Why must electricity production and consumption remain balanced?
2. Distinguish power from stored energy.
3. Why are batteries suitable for rapid grid services?
4. Explain the operating cycle of a pumped-storage plant.
5. Why can hydrogen be useful despite its lower efficiency?
6. Which criteria should engineers consider besides efficiency?

Engineering Challenge

A remote island has a 12 MW peak demand, a 20 MW wind farm and a 10 MW solar farm. Propose a storage strategy for frequency control, night-time supply and a five-day period of weak wind. Select at least two technologies and justify their power, capacity, duration, efficiency, safety and environmental impact.

Application Exercise

A pumped-storage plant transfers 5.0×10^6 m^3 of water through a vertical height of 420 m. Estimate the stored gravitational energy using $E = \rho V g h$, with $\rho = 1000\ \mathrm{kg\,m^{-3}}$. Express the result in GWh, then estimate the recoverable energy for a round-trip efficiency of 78%.

Speaking Task

In teams, defend one storage technology in a three-minute technical pitch. Then identify one service for which your technology would be a poor choice.

Sources

IEA – Grid-scale storage: online resource

NREL – Energy storage: online resource

EC06 – Electric Vehicles

Are electric cars really sustainable?

Learning Objectives

- Describe the main components of a battery-electric vehicle powertrain.
- Explain charging, regenerative braking and battery thermal management.
- Assess an electric-vehicle project using technical and environmental criteria.

Engineering News

Electric mobility is moving beyond the vehicle itself. Engineers now design complete systems that connect cars, charging stations, buildings and electricity networks. Smart charging can shift demand away from peak hours, while vehicle-to-grid operation may allow parked vehicles to provide flexibility to the power system.

Did You Know?

An electric motor can reverse its energy flow in a fraction of a second: it drives the wheels during acceleration and becomes a generator during regenerative braking.

Main Article

A battery-electric vehicle stores electrical energy in a high-voltage battery pack. During acceleration, the battery supplies direct current to a power-electronics inverter. The inverter controls the alternating current delivered to the traction motor, which converts electrical energy into mechanical torque at the wheels. Unlike a combustion engine, an electric motor can produce high torque from very low speed and commonly operates with high efficiency over a broad range of conditions.

The battery is not simply a collection of cells. It also includes electrical connections, contactors, sensors, cooling channels, structural protection and a **Battery Management System** (BMS). The BMS estimates state of charge and state of health, balances the cells and protects the pack against excessive current, voltage and temperature. Thermal management is critical because low temperatures reduce available power, while high temperatures accelerate ageing and may increase safety risks.

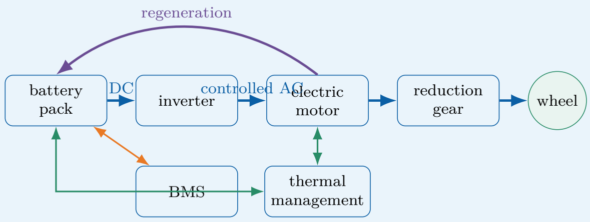
When the driver brakes, the inverter can operate the traction motor as a generator. Part of the vehicle's kinetic energy is then converted back into electrical energy and stored in the battery. This **regenerative braking** reduces energy consumption and mechanical-brake wear, although friction brakes remain necessary for emergency stops, low-speed operation and situations in which the battery cannot accept additional power.

Charging power depends on both the charging station and the vehicle. An on-board charger converts alternating current from a domestic or public AC supply. Fast DC charging bypasses the on-board charger and feeds controlled direct current to the battery. High charging power shortens stops, but it also increases heat generation, infrastructure cost and stress on the local electricity network.

Electric vehicles have no exhaust emissions during use, but they are not impact-free. Engineers must consider battery manufacture, electricity generation, vehicle mass, tyre particles, lifetime and recycling. Their environmental performance therefore depends on the complete life cycle and on how the electricity used for charging is produced.

Engineering Focus

Energy flow in a battery-electric vehicle

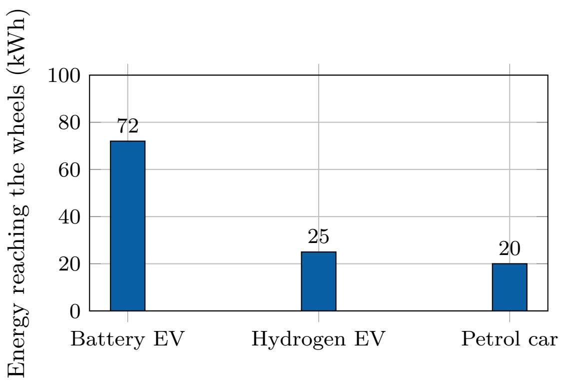


The high-voltage traction circuit is complemented by a DC/DC converter that supplies the vehicle's low-voltage network. Isolation monitoring and contactors disconnect the battery if a serious fault is detected.

Key Figures

Electric motor peak efficiency	90–97%
Typical battery voltage	350–800 V
Home AC charging	2–11 kW
Rapid DC charging	50–350 kW
Typical passenger-car battery	40–100 kWh

From supplied energy to the wheels



Illustrative conversion chain starting from 100 kWh of supplied energy. Actual values vary with technology, driving conditions and system boundaries.

Useful Vocabulary

battery pack	pack batterie
state of charge	état de charge
state of health	état de santé
traction motor	moteur de traction
regenerative braking	freinage régénératif
on-board charger	chargeur embarqué
charging point	point de recharge
power electronics	électronique de puissance

Questions

1. What are the functions of the inverter and the traction motor?
2. Why does a battery pack require thermal management?
3. Explain how regenerative braking changes the direction of energy flow.
4. Compare AC charging with rapid DC charging.
5. Why can charging power be limited by the vehicle rather than the station?
6. Which life-cycle impacts must be considered beyond exhaust emissions?
7. Why is smart charging useful for the electricity network?

Engineering Challenge

A city operates 24 buses, each travelling 220 km per day. Compare overnight depot charging with high-power opportunity charging at selected terminal stops. Propose a battery capacity, charging power and operating schedule. Justify your choices using range, charging time, battery ageing, network connection, redundancy, safety and maintenance.

Application Exercise

An electric bus has a usable battery capacity of 320 kWh and consumes on average 1.25 kWh/km. Estimate its theoretical range. Then calculate the energy recovered during a 12 km downhill section if the mechanical energy available at the wheels is 18 kWh and the regenerative chain efficiency is 72%. Explain why the practical operating range must remain lower than the theoretical value.

Speaking Task

Prepare a four-minute technical briefing answering the question: "Should all new urban cars be electric by 2035?" Present one technical advantage, one limitation and one policy measure that could improve the overall transport system.

Sources

IEA – Global EV Outlook: online resource

U.S. Department of Energy – Electric vehicles: online resource

Alternative Fuels Data Center – How electric vehicles work: online resource

EC07 – Robotics

How can robots work safely with humans?

Learning Objectives

- Identify the mechanical, electrical and software subsystems of a robot.
- Explain position, force and vision feedback in a collaborative workstation.
- Propose a safe robotic solution for an industrial task.

Engineering News

Collaborative robots, or **cobots**, are changing the organisation of factories and laboratories. Instead of operating permanently behind a fence, they can share selected tasks with people. The engineering objective is not merely to make the robot slower: the complete workstation must detect hazards, limit energy and bring the system to a safe state when required.

Did You Know?

A collaborative robot is not automatically a safe collaborative application. The gripper, the workpiece, the programmed speed and the surrounding equipment must all be included in the risk assessment.

Main Article

A robot is a programmable machine that senses its environment, processes information and acts through a mechanical structure. A typical articulated robot contains rigid links connected by joints. Electric servomotors create torque at these joints, while encoders measure angular position and speed. A controller compares the measured motion with the requested trajectory and continuously corrects the motor commands.

The tool attached to the final link is called the **end-effector**. It may be a gripper, a welding torch, a screwdriver or a camera. Selecting this tool requires engineers to consider payload, accuracy, cycle time, gripping force and the geometry of the product. The useful region that the end-effector can reach is the robot's workspace.

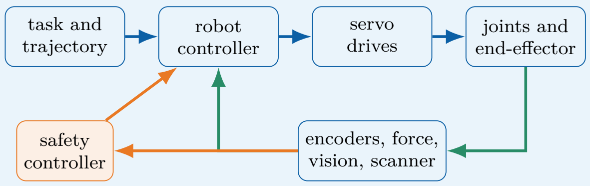
Position control is sufficient for many repetitive tasks, but physical interaction requires additional information. A force–torque sensor can detect contact, while motor-current monitoring can estimate resistance at a joint. Machine-vision systems locate parts, inspect surfaces and track people. These measurements form feedback loops that allow the robot to adapt rather than repeat a completely fixed sequence.

Safe collaboration depends on the entire application. Possible strategies include monitored stops, hand-guiding, speed and separation monitoring, and power-and-force limitation. A scanner or camera can define protective zones around the robot. As a person approaches, the controller first reduces speed and then stops the hazardous motion. Mechanical design also matters: rounded surfaces, low moving mass and the absence of crushing points can reduce injury risk.

Mobile robots combine localisation, mapping and path planning. Because sensors can fail or be obstructed, their design needs diagnostics and a defined safe response. Cybersecurity is equally important: access control, validated software and network segmentation help prevent dangerous commands or corrupted sensor data.

Engineering Focus

Closed-loop architecture of a collaborative robot

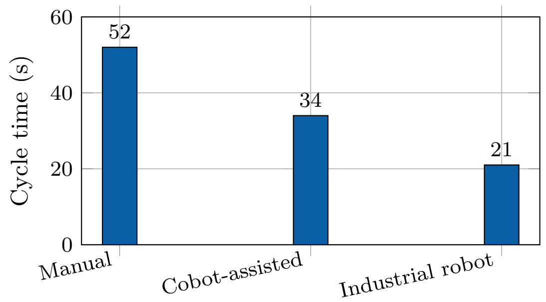


The standard motion controller aims to achieve the task. The safety controller independently supervises hazardous conditions and can request a controlled stop or remove drive power.

Key Figures

Typical articulated cobot axes	6–7
Repeatability of small industrial robots	about 0.02–0.1 mm
Typical cobot payload range	3–35 kg
Main feedback period	milliseconds
Emergency response objective	safe state

Assembly cycle comparison



Illustrative values for one assembly operation. A faster cycle does not by itself prove that a solution is safer, more flexible or more economical.

Useful Vocabulary

joint	articulation
link	solide / bras
actuator	actionneur
encoder	codeur
end-effector	effecteur terminal
payload	charge utile
repeatability	répétabilité
protective stop	arrêt de protection

Questions

1. What information is provided by an encoder?
2. Distinguish accuracy from repeatability.
3. Why may position control be insufficient during contact tasks?
4. Explain speed and separation monitoring.
5. Why must the end-effector be included in the risk assessment?
6. Give two possible consequences of a cybersecurity failure.
7. Why is productivity data alone insufficient when selecting a robot?

Engineering Challenge

A company wants to automate the insertion and tightening of four screws on a 2.5 kg product while an operator performs the adjacent wiring task. Design the collaborative workstation. Select the robot, end-effector and sensors; define the work sequence and protective zones; identify crushing, impact, tool and cybersecurity hazards; and propose validation tests. Compare the solution with a fully manual station and a fenced industrial robot.

Application Exercise

A collaborative robot moves a 4.0 kg component with a horizontal acceleration of 1.8 m/s^2. Calculate the resulting inertial force. If the end-effector has a mass of 1.5 kg, repeat the calculation for the complete moving load. Explain why this simple calculation is not sufficient to validate human–robot safety.

Speaking Task

Prepare a four-minute design review answering the question: "Should this workstation use a cobot or a conventional industrial robot?" Defend your choice using safety, cycle time, flexibility, payload, maintenance and cost.

Sources

International Federation of Robotics – industrial and service robots: online resource

NIST – robotics and autonomous systems: online resource

European Commission – machinery safety: online resource

EC08 – Artificial Intelligence

Can machines make reliable engineering decisions?

Learning Objectives

- Distinguish rule-based programming, machine learning and inference.
- Explain how data quality affects classification performance.
- Propose a validated AI architecture for an engineering application.

Engineering News

Artificial intelligence is now embedded in industrial inspection, predictive maintenance, energy management, medical imaging and autonomous systems. Its engineering value comes from its ability to process large quantities of sensor data, but performance figures alone are not enough: reliability, traceability and human responsibility must also be designed.

Did You Know?

An AI model can achieve a high overall accuracy while still missing most rare defects. When the classes are unbalanced, precision and recall are often more informative than accuracy alone.

Main Article

Traditional software follows explicit rules written by a programmer. A machine-learning system instead learns a mathematical model from examples. During **training**, an algorithm adjusts internal parameters so that the model's outputs approach the expected answers contained in a labelled dataset. During **inference**, the trained model processes new data and produces a classification, numerical estimate or recommended action.

In computer vision, a neural network can learn to distinguish acceptable products from defective ones. The input may be an image of a machined surface, while the output is a defect category and a confidence score. In predictive maintenance, vibration, temperature and current measurements can be used to estimate the condition of a bearing or motor. In both cases, the model is only one part of the complete engineering system: sensors, lighting, communication, computing hardware and the human-machine interface also determine the final result.

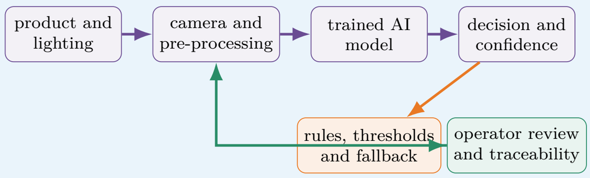
Dataset quality is critical. If defects are rare, poorly labelled or photographed under only one lighting condition, the model may appear accurate while failing in production. Engineers therefore separate data into training, validation and test sets. The test set must remain independent so that it measures generalisation rather than memorisation. Useful indicators include precision, recall, false-alarm rate and missed-detection rate.

Safety-critical applications require additional safeguards. A confidence threshold can prevent uncertain outputs from triggering an automatic decision. Independent physical checks, plausibility rules and human confirmation can also reduce risk. Engineers must monitor **data drift**: over time, new products, sensor ageing or environmental changes may make the operating data different from the original training data.

Explainability and traceability are also important. The development team should record dataset versions, model parameters, test results and software updates. If the system influences a hazardous action, responsibility remains with the organisation that designed and validated the complete system, not with the algorithm itself.

Engineering Focus

Validated AI inspection architecture

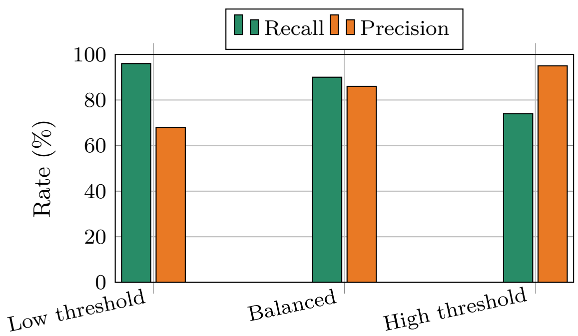


The AI model proposes a result. Independent thresholds and fallback rules determine whether the result may be accepted automatically or must be reviewed by a person. Production data and operator feedback are stored so that performance can be monitored over time.

Key Figures

Training set	model fitting
Validation set	parameter selection
Test set	final independent evaluation
False negative	missed defect
False positive	false alarm
Confidence threshold	application-dependent

Defect-detection trade-off



Illustrative values. Lowering the threshold detects more defects but may also create more false alarms. The correct setting depends on the consequences of each error.

Useful Vocabulary

dataset	jeu de données
training	entraînement
inference	inférence
label	étiquette / classe
accuracy	exactitude globale
precision	précision positive
recall	taux de détection
data drift	dérive des données

Questions

1. Distinguish training from inference.
2. Why must a test set remain independent?
3. Explain the difference between a false positive and a false negative.
4. Why can overall accuracy be misleading for rare defects?
5. What is the purpose of a confidence threshold?
6. Give two possible causes of data drift.
7. Why must responsibility remain with the engineering organisation?

Engineering Challenge

A factory wants to inspect 30 aluminium parts per minute using a camera and an AI model. A missed crack may lead to a dangerous failure, while a false alarm sends a good part for manual inspection. Design the complete inspection station. Specify the camera and lighting arrangement, dataset, performance indicators, acceptance threshold, fallback strategy, traceability records and validation tests. Explain how you would monitor the system after deployment.

Application Exercise

An image-inspection system processes 240 parts per minute. Its false-rejection rate is 1.5% and its false-acceptance rate is 0.20%. During an eight-hour shift, estimate the number of acceptable parts wrongly rejected if 96% of production is actually conforming, and the number of defective parts wrongly accepted. State one operational consequence of each type of error.

Speaking Task

Prepare a four-minute design review answering the question: "Should this inspection system reject parts automatically?" Defend your position using safety, productivity, precision, recall, traceability and human supervision.

Sources

NIST – Artificial Intelligence: online resource

OECD – Artificial Intelligence: online resource

European Commission – Artificial Intelligence: online resource

EC09 – Space Engineering

How do engineers design systems for extreme environments?

Learning Objectives

- Identify the main subsystems of an artificial satellite.
- Explain how engineers manage launch loads, vacuum and temperature variations.
- Propose a coherent architecture for a small Earth-observation mission.

Engineering News

Reusable launch vehicles, commercial constellations and small satellites are changing access to space. Lower launch costs make new missions possible, but every spacecraft must still survive an extreme environment and operate for years without direct maintenance.

Did You Know?

In low Earth orbit, a satellite travels at about 7.8 $\mathrm{km\,s^{-1}}$ and completes one orbit in roughly 90 minutes. It may therefore pass from sunlight to eclipse many times each day.

Main Article

A spacecraft is designed around its **mission**. An Earth-observation satellite must carry a camera or radar payload, point it accurately towards the ground, store the collected data and send it to a ground station. Around this payload, engineers build a service platform containing the structure, power supply, thermal control, communication system, onboard computer and attitude-control system.

Launch is the first major challenge. The satellite experiences acceleration, vibration and intense acoustic pressure. Its structure must therefore be light but stiff, while sensitive components are mounted and tested so that they do not resonate dangerously. After separation from the launcher, the environment changes completely: there is almost no atmosphere, heat cannot be removed by convection and electronic components are exposed to radiation.

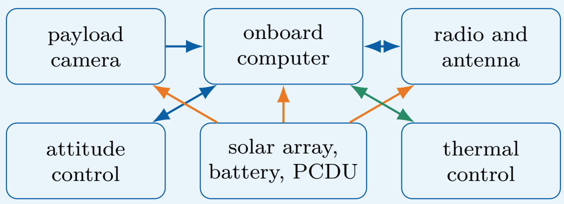
Electrical power usually comes from photovoltaic panels. Batteries supply the spacecraft during eclipse, when Earth blocks the Sun. A power-control unit regulates voltage and distributes energy to the payload, computer, heaters and transmitters. Because available power is limited, mission operations are planned carefully: a high-power transmitter or instrument may only be activated at specific times.

Thermal control is equally important. In sunlight, exposed surfaces may become very hot; in shadow, they may cool rapidly. Engineers use insulation blankets, reflective coatings, heat pipes, radiators and electrical heaters. The objective is not to keep every part at room temperature, but to keep each component within its qualified operating range.

A satellite must also control its orientation, called **attitude**. Sun sensors, star trackers and gyroscopes estimate its orientation. Reaction wheels rotate internally and make the spacecraft turn in the opposite direction. Small thrusters can desaturate the wheels or perform orbital corrections. Since repair is generally impossible, critical functions may be duplicated and the onboard software must detect faults and switch to a safe mode.

Engineering Focus

Typical small-satellite architecture

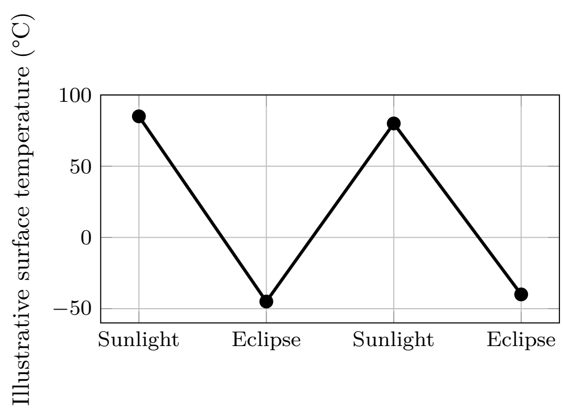


The payload performs the mission, while the platform keeps it powered, pointed, protected and connected. Interfaces between subsystems are as important as the components themselves.

Key Figures

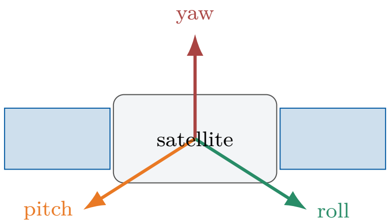
Typical low Earth orbit altitude	400–800 km
Orbital speed	about 7.8 km/s
One orbit	about 90 min
Vacuum pressure	extremely low
Repair after launch	usually impossible

Thermal cycle in orbit



Illustrative values only. Real temperatures depend on orbit, surface coating, orientation, internal dissipation and thermal-control design.

Three-axis attitude control



Sensors estimate orientation; reaction wheels or thrusters create the required rotation around the three axes.

Useful Vocabulary

spacecraft	engin spatial
payload	charge utile
launch load	effort au lancement
solar array	panneau solaire
eclipse	passage dans l'ombre
attitude	orientation
reaction wheel	roue de réaction
star tracker	viseur d'étoiles
thruster	propulseur
redundancy	redondance

Questions

1. Which mechanical loads affect a satellite during launch?
2. Why is convection unavailable in space?
3. What is the role of the battery during eclipse?
4. Name three devices used for attitude determination or control.
5. Why are critical functions sometimes duplicated?

Engineering Challenge

Design a 20 kg Earth-observation satellite. Draw a subsystem diagram and explain how the satellite will generate and store power, control its temperature, point its camera, communicate with Earth and react to one critical failure. Identify two design compromises involving mass, power, cost or reliability.

Application Exercise

A satellite in low Earth orbit receives 950 W from its solar arrays in sunlight and requires an average electrical power of 620 W. During a 36-minute eclipse, the battery supplies the complete load. Estimate the minimum energy that must be delivered by the battery in kWh. Then calculate the required nominal capacity if only 70% of the battery capacity may be used.

Speaking Task

Present your satellite in a four-minute mission review. Explain the mission, the architecture and the most serious technical risk. Conclude by defending one design compromise.

Sources

NASA Small Spacecraft Systems Virtual Institute: online resource

European Space Agency – Space Engineering and Technology: online resource

EC10 – Medical Engineering

How can technology improve healthcare?

Learning Objectives

- Describe the measurement chain of a connected medical device.
- Explain the roles of calibration, biocompatibility and risk analysis.
- Design a safe monitoring system combining sensors, software and human supervision.

Engineering News

Wearable sensors, robotic surgery, smart prostheses and remote monitoring are transforming healthcare. These systems can improve diagnosis and quality of life, but they must meet demanding requirements for safety, reliability, privacy and clinical usefulness.

Did You Know?

A device can be precise without being accurate. Repeated measurements may be very similar to one another but all shifted away from the true value. Calibration is used to identify and correct this systematic error.

Main Article

Medical engineering combines electronics, mechanics, materials science, computer science and human physiology. A medical device may simply measure a quantity, such as body temperature, or it may act on the patient, as a pacemaker, insulin pump or powered prosthesis does. In every case, the design must begin with a clearly defined medical need and a careful analysis of possible hazards.

Most monitoring devices contain a measurement chain. A sensor converts a physiological quantity into an electrical signal. Because this signal is often weak and noisy, analogue electronics amplify and filter it before an analogue-to-digital converter sends numerical data to a processor. Software then calculates indicators, detects abnormal values and displays information to a patient or healthcare professional. Calibration links the electrical output to a trustworthy physical measurement.

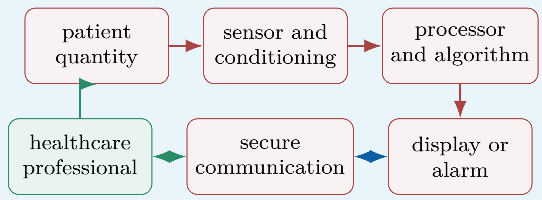
A pulse oximeter, for example, shines red and infrared light through a finger and measures how much light is absorbed. From the changing optical signals, the device estimates blood oxygen saturation and pulse rate. Movement, poor contact or ambient light can disturb the measurement, so the system must detect low-quality signals rather than display a misleading result.

Implantable devices create additional constraints. Their materials must be biocompatible, their energy consumption must be extremely low and their enclosure must protect the electronics from body fluids. A pacemaker continuously monitors cardiac activity and delivers a controlled electrical pulse only when required. This is a closed-loop system: measurement, decision and action are linked.

Connected devices can transmit measurements to a smartphone or hospital server. Remote monitoring may allow earlier intervention, but it also introduces cybersecurity and privacy risks. Engineers must protect communication, control software updates and ensure that a network failure cannot place the patient in danger. Clinical validation remains essential: a technically accurate sensor is only useful if its information supports an appropriate medical decision.

Engineering Focus

Connected patient-monitoring chain

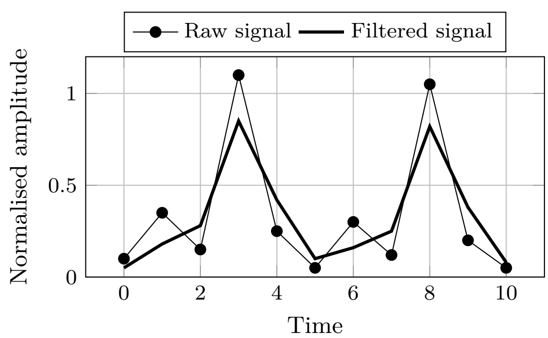


The device provides information, but the complete system includes the patient, the user interface, communication network and medical decision. Each interface must be validated.

Key Figures

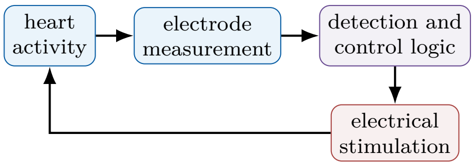
Sensor output	often millivolts or microvolts
Implant operating life	several years
Acceptable false alarm rate	application-dependent
Software traceability	required throughout development
Patient data	confidential

Filtering a physiological signal



Illustrative data. Filtering reduces rapid variations, but excessive filtering may also delay or distort clinically important information.

Closed-loop pacemaker principle



The controller stimulates the heart only when the measured rhythm does not satisfy the programmed criteria. A safe mode must remain available if a sensor or software fault is detected.

Useful Vocabulary

medical device	dispositif médical
implant	implant
prosthesis	prothèse
biocompatible	biocompatible
calibration	étalonnage
signal conditioning	conditionnement du signal
false alarm	fausse alarme
clinical validation	validation clinique
privacy	confidentialité
closed-loop control	commande en boucle fermée

Questions

1. Why must a weak sensor signal be amplified and filtered?
2. Distinguish precision from accuracy.
3. Give two possible disturbances affecting a pulse oximeter.
4. Why must an implant consume very little energy?
5. What makes a pacemaker a closed-loop system?

Engineering Challenge

Design a connected device that monitors an elderly patient at home. Select two physiological or activity measurements, draw the information chain and define alarm conditions. Explain how the system detects poor-quality data, protects privacy, continues operating during a network failure and keeps a healthcare professional in the decision loop.

Application Exercise

A physiological sensor produces 2.4 mV for the measured condition. The acquisition system requires a 1.8 V signal at the analogue-to-digital converter input. Calculate the ideal amplifier gain. If an offset of 15 mV is also amplified, calculate its contribution at the output and explain why offset correction is necessary.

Speaking Task

Should a connected medical device automatically contact emergency services when it detects a critical event? Present the benefits, risks and safeguards in a four-minute design review.

Sources

World Health Organization – Medical Devices: online resource

U.S. Food and Drug Administration – Medical Devices: online resource

EC11 – Additive Manufacturing

Can 3D printing transform industrial production?

Learning Objectives

- Explain the digital chain from CAD model to finished part.
- Compare additive manufacturing with machining and casting.
- Identify the main design rules, benefits and limitations of metal 3D printing.

Engineering News

Additive manufacturing is now used for rocket engines, aircraft components, medical implants and industrial tooling. Its main advantage is not simply that it prints parts: it allows engineers to produce geometries that conventional processes cannot easily manufacture.

Main Article

Additive manufacturing creates a component layer by layer from a digital model. The process begins with a three-dimensional CAD file. Slicing software divides the geometry into thin horizontal sections and generates the machine instructions. Depending on the technology, a nozzle deposits a molten polymer, a laser melts metal powder, or light solidifies a liquid resin.

This manufacturing route changes the designer's freedom. Internal cooling channels, lattice structures and organic shapes can be integrated into a single component. Topology optimisation can remove material from low-stress zones while preserving load paths. In aerospace engineering, this can reduce mass and therefore fuel consumption. In medicine, a porous implant can be adapted to a patient and encourage bone growth.

The process also has constraints. Overhanging zones may require temporary supports. These supports consume material and must be removed after printing. Thermal gradients can create residual stress, warping or cracks, especially in metal parts. Build orientation affects surface quality, printing time and mechanical properties. The engineer must therefore design both the part and its manufacturing strategy.

A printed component is rarely ready for immediate use. Powder removal, heat treatment, support removal, machining and inspection may be required. Non-destructive testing can detect internal porosity, while dimensional measurement verifies that the geometry is within tolerance. Because mechanical properties may vary with the build direction, qualification is essential for safety-critical parts.

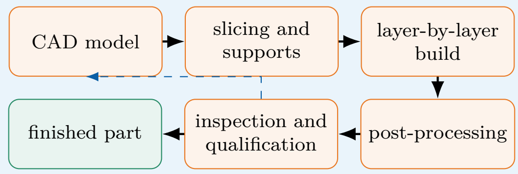
Additive manufacturing is particularly effective for prototypes, customised products, spare parts and low-volume high-value components. It is less competitive for millions of simple identical parts. The best industrial solution is often hybrid: printing creates the complex near-net shape, then conventional machining produces accurate interfaces and smooth functional surfaces.

Did You Know?

A component redesigned for additive manufacturing can combine several conventional parts into one. This reduces assembly operations, fasteners and possible leakage points, but it can make inspection and repair more difficult.

Engineering Focus

Digital manufacturing chain

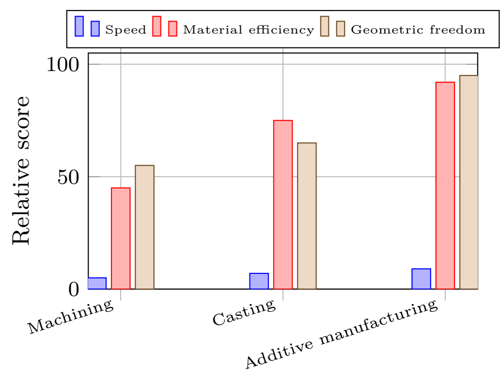


Inspection may lead to a design or parameter change. The chain is therefore iterative rather than strictly linear.

Key Figures

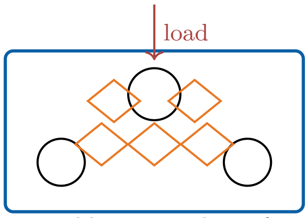
Typical polymer layer	0.10–0.30 mm
Typical metal layer	0.02–0.08 mm
Main input	3D CAD model
Frequent post-processes	heat treatment, machining
Best production range	low volume / high value

Comparing manufacturing routes



Illustrative comparison. The most suitable process depends on quantity, material, tolerance, geometry and total lifecycle cost.

Designing a lightweight bracket



material kept around interfaces and principal load paths

Topology optimisation must be followed by checks on minimum wall thickness, support removal, powder evacuation and machinable reference surfaces.

Useful Vocabulary

additive manufacturing	fabrication additive
slicing	découpage en couches
powder bed	lit de poudre
nozzle	buse
support structure	structure de support
build orientation	orientation de fabrication
warping	déformation / gauchissement
porosity	porosité
post-processing	post-traitement
topology optimisation	optimisation topologique

Questions

1. What information does slicing software generate?
2. Why does build orientation influence a printed part?
3. Give two examples of post-processing operations.
4. Why can mechanical properties be anisotropic?
5. When is additive manufacturing more suitable than mass production?

Application Exercise

A metal part is 120 mm high and is printed with a layer thickness of 0.04 mm. Each layer takes 18 seconds to scan and recoating adds no extra time in this simplified model.

1. Calculate the number of layers.
2. Estimate the ideal build time in hours.
3. Add 25% for calibration and interruptions. Estimate the total production time.

Engineering Challenge

Redesign an aluminium bracket for metal additive manufacturing. Define the loads and interfaces, propose a lighter geometry and select a build orientation. Identify the required supports, two inspection methods and the surfaces to machine after printing.

Speaking Task

Will additive manufacturing replace conventional factories? Give a balanced three-minute answer with examples of suitable and unsuitable products.

Sources

NIST – Additive Manufacturing: online resource
NASA – Additive Manufacturing: online resource

EC12 – Smart Materials

Can materials react to their environment?

Learning Objectives

- Identify the stimulus and response of several smart materials.
- Explain how shape-memory and piezoelectric effects are used in engineering systems.
- Evaluate performance, fatigue, control and lifecycle constraints.

Engineering News

Smart materials can sense or respond to temperature, stress, electricity, light or magnetic fields. They are used in medical devices, vibration control, precision positioning and energy-efficient buildings, where one compact material can perform both structural and functional roles.

Main Article

A conventional material is selected mainly for passive properties such as stiffness, strength or thermal conductivity. A smart material produces a predictable response to an external stimulus. The response may be a change in shape, electrical charge, colour, viscosity or magnetic behaviour. Engineers use this coupling to create sensors, actuators and adaptive structures.

Shape-memory alloys are a well-known example. Nickel-titanium can be deformed in a low-temperature phase and recover a previously defined shape when heated. This effect results from a reversible change in crystal structure. A thin wire can therefore act as a compact actuator: electrical current heats it, the wire contracts, and a spring or external load returns it when it cools. Applications include medical stents, valves and lightweight mechanisms.

Piezoelectric materials couple mechanical and electrical energy. When compressed or bent, they produce electrical charge; this is the direct effect used in force, pressure and vibration sensors. When voltage is applied, they deform slightly; this inverse effect is used in ultrasound transducers, fuel injectors and nanometre-scale positioning systems. Their movement is small, but it is rapid and precise.

Other smart materials provide different functions. Electrochromic layers change optical transmission when a voltage is applied, allowing windows to control solar gain. Magnetorheological fluids become more resistant to flow in a magnetic field and can adjust the damping of a suspension. Self-healing polymers may release a repair agent when a crack ruptures embedded microcapsules.

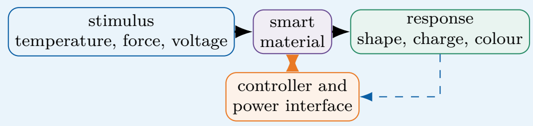
These materials are not intelligent by themselves. A useful engineering system still requires sensors, power electronics, control logic and safe limits. Performance can change with temperature, fatigue and ageing. The designer must also consider response time, energy consumption, recyclability and what happens when the active effect fails. A passive safe state is often essential.

Did You Know?

A piezoelectric element can be used as both a sensor and an actuator. The same physical coupling converts mechanical energy into electrical energy and electrical energy into mechanical motion.

Engineering Focus

Stimulus–material–response chain

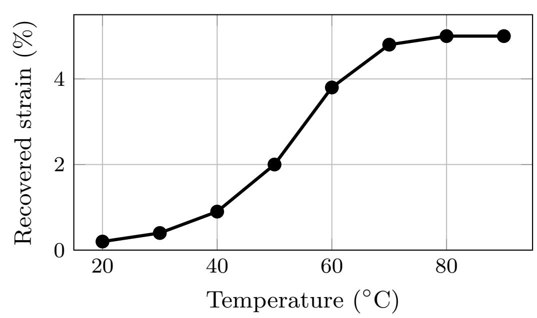


Closed-loop control measures the actual response and adjusts the stimulus. This improves accuracy and compensates for temperature variation or ageing.

Key Figures

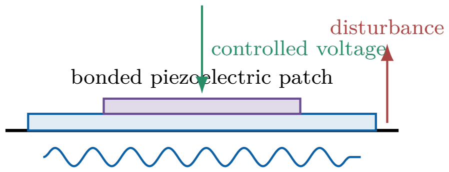
Shape-memory alloy response	several percent strain
Piezoelectric actuator motion	micrometres
Piezoelectric response	very fast
Electrochromic transition	seconds to minutes
Critical design issue	fatigue and ageing

Shape recovery with temperature



Illustrative response of a shape-memory actuator. The transformation occurs over a temperature range rather than at one exact temperature.

Piezoelectric vibration control



structural vibration measured and reduced by an opposing actuator force

A sensor measures vibration, the controller calculates a corrective command and the piezoelectric patch produces a small deformation that opposes the motion.

Useful Vocabulary

smart material	matériau intelligent
stimulus	sollicitation / stimulus
shape-memory alloy	alliage à mémoire de forme
phase transformation	transformation de phase
piezoelectric effect	effet piézoélectrique
strain	déformation relative
damping	amortissement
self-healing	autoréparant
ageing	vieillissement
fail-safe state	état sûr en cas de panne

Questions

1. What distinguishes a smart material from a passive material?
2. How can a shape-memory wire act as an actuator?
3. Explain the direct and inverse piezoelectric effects.
4. Why is closed-loop control useful with smart materials?
5. Give two limitations that must be considered during design.

Application Exercise

A piezoelectric actuator is 40 mm long and produces a free strain of 0.08% when powered.

1. Calculate its free displacement in millimetres and micrometres.
2. A mechanical amplifier multiplies the displacement by 4. Estimate the output movement.
3. Explain why the real movement will generally be lower than this ideal value.

Engineering Challenge

Design an adaptive bicycle helmet ventilation system using a smart material. Define the stimulus, material response and mechanical function. Add a simple control strategy, estimate the required movement and explain how the helmet remains safe if the active material or power supply fails.

Speaking Task

Choose one smart material and present an everyday product that could use it. Explain the benefit, technical limitation and environmental impact in a three-minute product pitch.

Sources

NIST – Materials Research: online resource

European Space Agency – Space Engineering and Technology: online resource

EC13 – Smart Cities

Can connected systems make cities more sustainable?

Learning Objectives

- Describe the architecture of a connected urban service.
- Explain how sensing, communication and control can reduce resource consumption.
- Discuss interoperability, privacy, resilience and social inclusion.

Engineering News

Cities are installing connected lighting, traffic and environmental-monitoring systems. The most successful projects do not simply collect data: they convert measurements into useful decisions while protecting citizens and keeping essential services available during failures.

Main Article

A smart city uses sensors, communication networks and software to improve urban services. A typical system begins with distributed devices that measure traffic flow, air quality, noise, temperature, water level or electricity consumption. Data are transmitted to a local controller or digital platform, where algorithms detect trends, identify abnormal conditions and select an appropriate action.

Street lighting provides a simple example. Conventional lamps operate at full power for most of the night. An adaptive system can combine a light sensor, a motion detector and a timetable. The lamps dim when streets are empty and return to a safe level when a pedestrian or vehicle is detected. Energy savings are possible, but minimum illumination, reaction time and failure behaviour must be specified carefully.

Traffic management is more complex. Cameras, induction loops and connected vehicles estimate queue lengths. A controller can modify signal timing to reduce waiting time and emissions. The local optimisation of one junction, however, may create congestion elsewhere. Engineers therefore model the complete network and define priorities for buses, emergency vehicles, cyclists and pedestrians.

Smart buildings and smart grids can also cooperate. Occupancy sensors adjust heating and ventilation, while a district energy manager coordinates photovoltaic generation, batteries and electric-vehicle charging. This requires devices from different manufacturers to exchange data. Interoperability and open communication standards are therefore essential. Edge computing can also process urgent data close to the sensors, reducing latency and dependence on a distant platform.

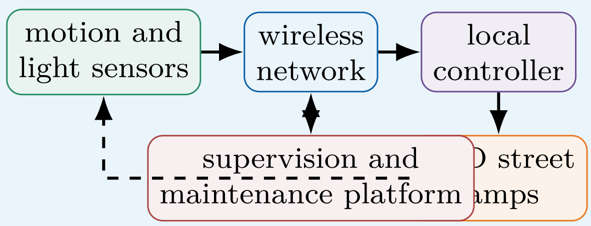
Connectivity also creates risks. Location or camera data may reveal personal behaviour. A network failure must not stop traffic lights, water pumps or emergency systems. Good design separates critical functions, stores only necessary data, applies cybersecurity measures and includes redundant local fallback modes. Finally, a city is not truly smart if services are inaccessible to people without smartphones or digital skills.

Did You Know?

A digital twin is a computer model linked to measurements from the real city. Engineers can use it to compare traffic, energy or flood-management scenarios before changing physical infrastructure.

Engineering Focus

Architecture of an adaptive lighting service

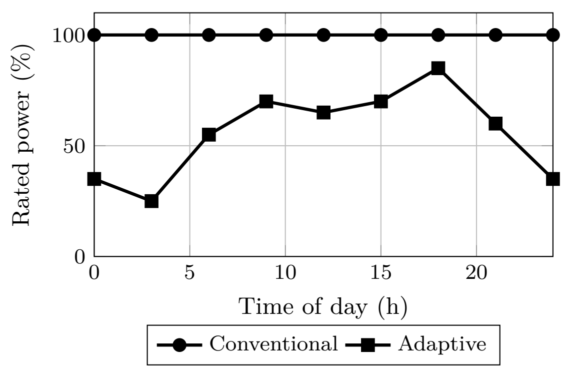


The essential control loop is local: sensing, decision and lighting remain available even if the central platform is disconnected.

Key Figures

Connected devices	thousands per district
Useful lamp response time	a few seconds
Important design property	interoperability
Critical service requirement	fallback mode
Main ethical concern	privacy

Lighting power profile



Illustrative daily profiles. The adaptive system lowers the power demand during quiet periods while restoring a higher level when activity is detected.

Useful Vocabulary

sensor network	réseau de capteurs
traffic flow	flux de circulation
street lighting	éclairage public
occupancy	occupation / présence
interoperability	interopérabilité
digital twin	jumeau numérique
fallback mode	mode de repli
outage	panne / interruption
privacy	vie privée
urban planning	urbanisme

Questions

1. Describe the data path in a smart-city system.
2. How can adaptive lighting save energy without reducing safety?
3. Why can local traffic optimisation be insufficient?
4. What does interoperability mean in this context?
5. Give two ways to make a connected urban service resilient.

Application Exercise

A district operates 1,200 LED street lamps rated at 80 W. The conventional system runs at full power for 12 hours each night. An adaptive strategy uses full power for 5 hours and 35% power for the remaining 7 hours.

1. Calculate the nightly energy consumption of the conventional system.
2. Calculate the nightly energy consumption of the adaptive system.
3. Determine the energy saved per night and the percentage reduction.

Engineering Challenge

Design a smart energy-management system for a school district. Select sensors and controlled equipment, draw the information flow, define one energy-saving rule and describe the safe fallback mode. Explain how personal data will be minimised.

Speaking Task

Should a city use cameras and connected sensors to reduce congestion? Present one technical benefit, one privacy risk and one condition for public acceptance.

Sources

European Commission – Smart Cities: online resource

NIST – Smart Connected Systems: online resource

EC14 – Water Treatment

How can engineering provide safe drinking water?

Learning Objectives

- Identify the principal stages of drinking-water treatment.
- Explain filtration, disinfection and reverse osmosis.
- Select a robust treatment chain from water-quality and local constraints.

Engineering News

Membrane filtration, ultraviolet disinfection and low-energy desalination are improving access to safe water. Yet the best solution is not always the most advanced one: reliability, operator skills, spare parts and energy availability often determine whether a plant succeeds.

Main Article

Raw water may contain suspended particles, microorganisms, organic matter, dissolved salts and chemical pollutants. No single process removes every contaminant, so drinking-water plants combine several stages. The treatment chain is selected according to the source-water quality, local constraints and the required quality of the final water.

Screens first remove leaves and large debris. During coagulation, a chemical reagent destabilises very small particles that would otherwise remain in suspension. Gentle mixing then creates larger flocs. In a sedimentation tank, these flocs settle under gravity and form sludge that must be collected and treated.

Filtration removes the particles that remain. Sand filters trap material inside a porous bed, whereas activated carbon also adsorbs some organic molecules responsible for taste, odour or pollution. Membrane systems use pores of controlled size. As the membrane becomes dirty, pressure loss increases, so periodic backwashing or chemical cleaning is necessary.

Disinfection inactivates harmful microorganisms. Chlorine can maintain protection throughout the distribution network, whereas ultraviolet light acts rapidly but leaves no disinfectant residual. Engineers often combine methods and monitor turbidity, pH, conductivity and disinfectant concentration continuously.

Reverse osmosis is used for desalination. Pumps apply a pressure greater than the natural osmotic pressure, forcing water through a semi-permeable membrane while most salts remain in a concentrated brine. The process produces high-quality water but requires energy, pretreatment and careful brine management.

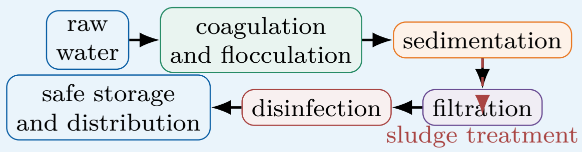
For a remote community, maintainability may be more important than maximum theoretical performance. Gravity flow, modular filters, solar pumping and simple sensors can reduce operating costs. A robust system must also include safe storage and prevent treated water from being contaminated again.

Did You Know?

Clear-looking water is not necessarily safe. Pathogenic microorganisms are invisible, so treatment performance must be verified by measurements and microbiological analysis.

Engineering Focus

Multi-barrier treatment chain

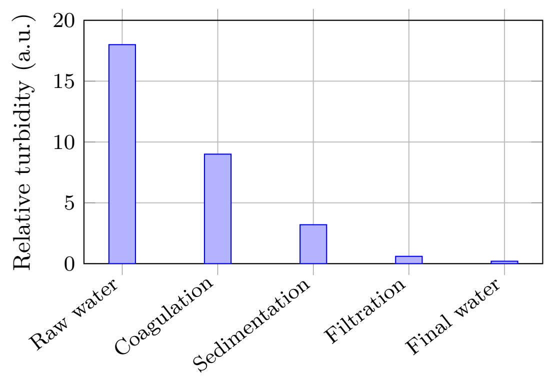


Each stage is a barrier. If one process performs less effectively, the remaining barriers reduce the risk of unsafe water reaching consumers.

Key Figures

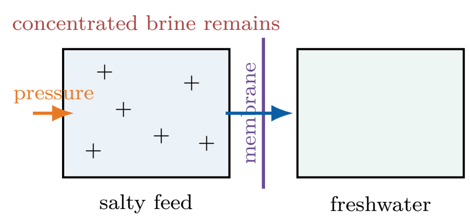
Typical raw-water turbidity	highly variable
Filtered-water target	very low turbidity
Reverse-osmosis driving force	high pressure
Main maintenance issue	fouling
Essential final step	safe storage

Turbidity through the treatment chain



Successive barriers progressively reduce suspended matter. The values are illustrative; actual performance depends on the source water and operating conditions.

Reverse osmosis principle



Useful Vocabulary

raw water	eau brute
coagulation	coagulation
floc	floc
sedimentation	décantation
activated carbon	charbon actif
backwashing	lavage à contre-courant
disinfection	désinfection
turbidity	turbidité
reverse osmosis	osmose inverse
brine	saumure

Questions

1. Why does a plant use several treatment stages?
2. Explain the difference between coagulation and sedimentation.
3. What additional function can activated carbon provide?
4. Compare chlorine and ultraviolet disinfection.
5. Why are pretreatment and maintenance important in reverse osmosis?

Application Exercise

A small plant treats 500 m^3 of water per day. Raw water has a turbidity of 18 NTU. After sedimentation, turbidity is reduced by 75%, and the filter then removes 92% of the remaining turbidity.

1. Calculate the turbidity after sedimentation.
2. Calculate the turbidity after filtration.
3. A target of 0.5 NTU is required. State whether the simplified treatment chain meets it.

Engineering Challenge

Design a compact treatment system for a village beside a turbid river. Choose the treatment stages, energy source and monitoring sensors. Explain routine maintenance and the action taken when one measurement exceeds its safe limit.

Speaking Task

Compare a central treatment plant with small local treatment units. Which solution is more suitable for a remote region, and under which conditions?

Sources

World Health Organization – Drinking Water: online resource

U.S. Environmental Protection Agency – Drinking Water: online resource

EC15 – Future Transport

How will people and goods move tomorrow?

Learning Objectives

- Compare transport modes using energy, capacity and infrastructure criteria.
- Explain the perception–decision–action chain of an autonomous vehicle.
- Evaluate a multimodal solution rather than a single vehicle technology.

Engineering News

Battery buses, hydrogen trains, autonomous shuttles and electric aircraft are being tested around the world. The largest environmental gains, however, may come from combining efficient vehicles with public transport, active mobility and better use of existing infrastructure.

Main Article

Transport engineering must move people and goods safely, reliably and with limited energy and land use. Because journey length, passenger capacity and infrastructure differ, no propulsion technology is ideal for every application.

Battery-electric vehicles are efficient because electric motors convert a large proportion of the stored energy into motion and can recover energy during braking. They are well suited to urban buses, cars and delivery vehicles that return regularly to a charging point. Long-distance freight requires larger batteries, faster charging or another energy carrier. Hydrogen fuel cells may reduce vehicle mass for some heavy applications, but producing, compressing and converting hydrogen introduces additional energy losses.

Rail carries many passengers or large freight loads with low rolling resistance. Electrified trains can use energy directly from the grid and do not need to carry a large energy store. High-speed rail can replace some short-haul flights when stations connect city centres and total door-to-door time is competitive.

Autonomous vehicles add a digital engineering challenge. Cameras provide detailed images, radar measures distance and relative speed, and lidar creates a three-dimensional representation. Sensor fusion combines these signals. Software then identifies objects, predicts their motion, plans a safe trajectory and commands steering and braking. Adverse weather, unusual road layouts and human behaviour make validation difficult.

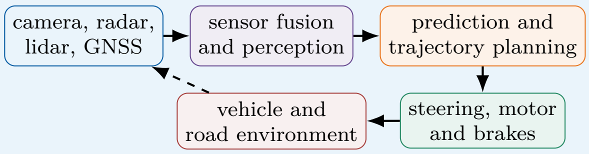
Future mobility is therefore a system-level problem. A journey may combine walking, bicycle, train and an on-demand shuttle. Ticketing, timetables and physical interchanges must work together. Engineers must also consider accessibility, affordability, charging networks, vehicle manufacture and the carbon intensity of the energy supply. Replacing every conventional car with another type of car would not, by itself, eliminate congestion or the demand for parking space.

Did You Know?

The energy efficiency of a vehicle is only one part of transport efficiency. Occupancy is equally important: a nearly empty heavy vehicle may consume more energy per passenger than a fuller one.

Engineering Focus

Autonomous driving control chain

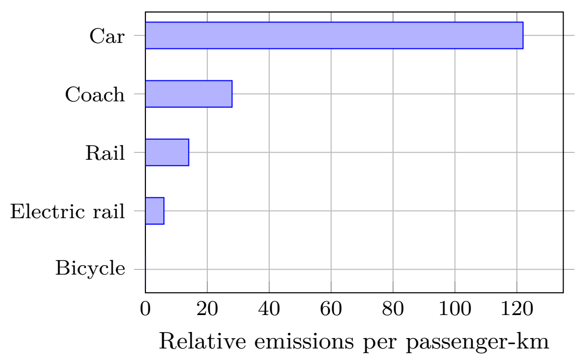


The loop must continue safely even when a sensor is degraded. Redundancy, plausibility checks and a minimum-risk manoeuvre are therefore central design features.

Key Figures

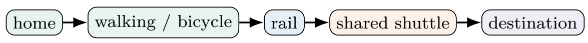
Electric motor efficiency	high
Rail rolling resistance	low
Autonomous control cycle	repeated continuously
Important system metric	passenger-km
Main urban constraint	limited space

Illustrative operational emissions



Illustrative comparison only. Actual emissions depend strongly on occupancy, vehicle, energy source, route and lifecycle boundaries.

Multimodal journey



A successful system minimises waiting, transfer distance and uncertainty between modes.

Useful Vocabulary

freight	fret
short-haul flight	vol court-courrier
rolling resistance	résistance au roulement
sensor fusion	fusion de capteurs
trajectory planning	planification de trajectoire
minimum-risk manoeuvre	manœuvre de risque minimal
shared mobility	mobilité partagée
interchange	pôle d'échanges
occupancy rate	taux de remplissage
passenger-kilometre	voyageur-kilomètre

Application Exercise

A class of 30 students travels 18 km to a science museum. Compare two options using the illustrative values in the graph: five cars carrying six students each, or one coach carrying all 30 students.

1. Calculate the total passenger-kilometres for the trip.
2. Estimate the operational emissions for each option in relative units.
3. By what factor does the coach reduce emissions compared with the cars?
4. Give two reasons why a real engineering comparison should also include lifecycle emissions.

Questions

1. Why is no propulsion technology suitable for every journey?
2. Give one advantage of electrified rail over battery vehicles.
3. Describe the autonomous perception–decision–action loop.
4. Why is vehicle occupancy important when comparing transport modes?
5. Which features make a multimodal journey attractive to users?

Engineering Challenge

Propose a low-carbon transport plan between a suburban school and a city centre. Compare at least three modes using capacity, journey time, energy, accessibility and infrastructure. Include one solution for the first and last kilometre.

Speaking Task

Should city centres restrict private cars? Defend a position using technical, environmental and social arguments, and propose one realistic alternative.

Sources

International Energy Agency – Transport: online resource

European Commission – Mobility and Transport: online resource

EC16 – Cybersecurity

How can engineers protect connected systems?

Learning Objectives

- Distinguish confidentiality, integrity and availability.
- Explain defence in depth for a connected engineering system.
- Analyse technical risk using threats, vulnerabilities and consequences.

Engineering News

Industrial systems, vehicles, hospitals and energy networks increasingly depend on connected software. Cybersecurity is therefore a safety and business-continuity issue as well as an information-technology issue: a digital attack can produce physical consequences.

Main Article

A connected engineering system contains sensors, controllers, software, communication links and actuators. Each interface can become an entry point for accidental failure or malicious action. Cybersecurity aims to preserve three central properties: confidentiality prevents unauthorised disclosure, integrity prevents unauthorised modification, and availability keeps the service usable.

Risk depends on more than the mere existence of a threat. Engineers identify valuable assets, possible attackers, vulnerabilities and consequences. A weak password on a non-critical display is not the same risk as an unprotected remote command to a water-treatment pump. Safety analysis and cyber-risk analysis must therefore be coordinated.

No single protection is sufficient. Defence in depth combines several independent layers. Devices should authenticate users and other devices. Communications can be encrypted, software should be updated, and networks should be segmented so that an office computer cannot directly control a critical machine. Access rights follow the principle of least privilege: each account receives only the permissions required for its task.

Monitoring is also necessary. Logs, alarms and anomaly detection can reveal unexpected traffic or commands. When an incident occurs, operators need a prepared response: isolate affected equipment, maintain a safe physical state, restore trusted software and verify operation. Offline backups help recovery, but they must be tested rather than merely stored.

Security creates engineering trade-offs. Encryption and authentication use processing time and energy. Updates may interrupt production, while legacy equipment may not support modern protocols. A secure design therefore begins with requirements and threat modelling rather than with a protective tool added at the end.

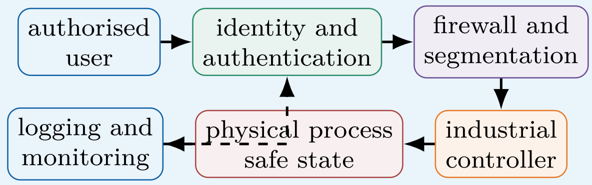
Human factors matter too: clear procedures and training reduce errors and social engineering attacks.

Did You Know?

Safety and cybersecurity can conflict. An emergency door should open quickly for evacuation, yet access control should prevent intrusion. The final design must satisfy both requirements.

Engineering Focus

Defence-in-depth architecture

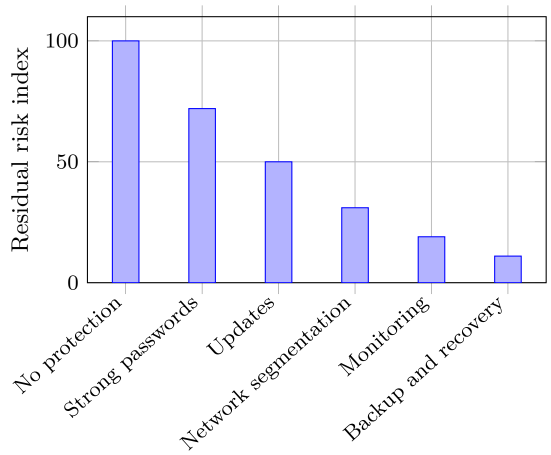


If one layer fails, another should still reduce the probability or consequence of compromise.

Key Figures

Core security objectives	3
Preferred access principle	least privilege
Critical architecture measure	segmentation
Recovery requirement	tested backup
Essential lifecycle activity	updates

Effect of layered safeguards



Illustrative risk reduction: safeguards are cumulative, but risk never becomes exactly zero and each layer must be maintained.

Risk reasoning

threat → vulnerability → unwanted event → consequence

Safeguards may reduce the likelihood of the event, limit its consequences, or improve detection and recovery.

Useful Vocabulary

confidentiality	confidentialité
integrity	intégrité
availability	disponibilité
threat	menace
vulnerability	vulnérabilité
authentication	authentification
encryption	chiffrement
network segmentation	segmentation du réseau
least privilege	moindre privilège
backup	sauvegarde

Application Exercise

The graph assigns a residual-risk index of 100 to an unprotected system and 11 after all listed safeguards have been introduced.

1. Calculate the percentage reduction in residual risk.
2. Which safeguard produces the largest absolute reduction between two consecutive bars?
3. Explain why the final index is not zero.
4. A greenhouse must continue watering crops during a network outage. Propose one local fail-safe mode and one method for recording the incident.

Questions

1. Define confidentiality, integrity and availability.
2. Why should cyber risk be linked to physical consequences?
3. Give four layers of a defence-in-depth strategy.
4. What is the purpose of network segmentation?
5. Why must backups and incident procedures be tested?

Engineering Challenge

Secure a connected greenhouse containing temperature sensors, an irrigation pump and remote access. Identify assets, one threat and one vulnerability. Propose authentication, segmentation, monitoring and a safe local mode if the network is unavailable.

Speaking Task

Is a fully connected system always better than an isolated one? Compare functionality, maintenance, security and resilience using one engineering example.

Sources

NIST – Cybersecurity Framework: online resource

European Union Agency for Cybersecurity: online resource

EC17 – Internet of Things

How can connected objects remain useful, secure and energy-efficient?

Learning Objectives

- Describe the architecture of an IoT system from sensing to user service.
- Compare communication technologies and explain the importance of energy management.
- Identify reliability, interoperability, privacy and cybersecurity requirements.

Engineering News

Connected sensors now monitor factories, buildings, farms and transport networks. The engineering challenge is no longer simply to connect a device, but to keep thousands of low-power objects useful, secure and maintainable throughout their entire service life.

Main Article

The Internet of Things (IoT) links physical objects to digital services. A typical object contains a sensor, a microcontroller, a communication interface and a power supply. It measures a physical quantity, processes the signal locally and sends useful information to a gateway or cloud platform. The platform stores data, detects abnormal behaviour and may send a command back to an actuator.

A connected temperature sensor illustrates the complete chain. The sensing element converts temperature into an electrical signal. The microcontroller filters the measurement, adds a timestamp and decides whether transmission is necessary. Sending every value provides detailed data but wastes energy and network capacity. Edge processing can instead transmit only a periodic summary or an alarm when a threshold is exceeded.

Communication technology depends on range, data rate, power and infrastructure. Bluetooth Low Energy is suited to short-range devices. Wi-Fi offers higher throughput but usually consumes more power. LoRaWAN can transmit small packets over several kilometres, while cellular IoT provides wide-area coverage through an operator network. No protocol is best for every application.

Energy autonomy is often the limiting requirement. Engineers reduce consumption by using sleep modes, low sampling rates and short radio transmissions. A battery-powered sensor may need to operate for several years without maintenance. Energy harvesting from light, vibration or heat can extend battery life, but the available power is small and variable.

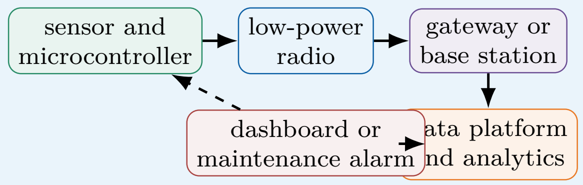
Security must be designed from the start. Each object needs a unique identity, authenticated software updates and encrypted communication. A compromised device can reveal private data or become an entry point into a larger system. Finally, interoperability matters: devices from different manufacturers must use compatible data formats so that the system can evolve without replacing every component.

Did You Know?

For a battery-powered object, radio transmission may consume far more energy than the measurement itself. Sending less data is therefore often more effective than improving the sensor electronics.

Engineering Focus

Architecture of a connected monitoring system

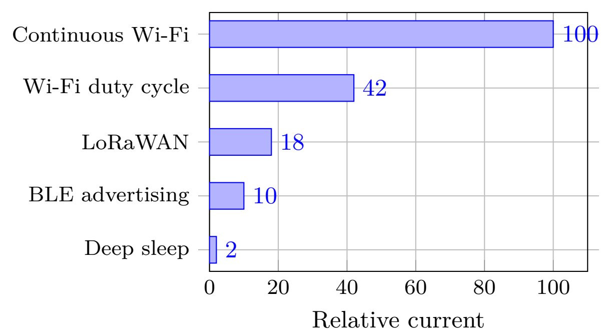


Local processing reduces traffic and allows basic operation even if the cloud connection is lost.

Key Figures

Typical sensor data rate	low
Critical resource	battery energy
Long-range low-power example	LoRaWAN
Short-range example	BLE
Essential maintenance feature	remote update

Relative communication current



Illustrative values: actual consumption depends strongly on hardware, coverage and transmission frequency.

Useful Vocabulary

connected object	objet connecté
gateway	passerelle
throughput	débit
sleep mode	mode veille
edge processing	traitement local
packet	paquet de données
energy harvesting	récupération d'énergie
firmware update	mise à jour du micrologiciel
interoperability	interopérabilité
encryption	chiffrement

Application Exercise

A connected sensor is powered by a 2400 mAh battery. During each hour, it remains in deep sleep for 3590 s at a relative current of 2, then transmits for 10 s using LoRaWAN at a relative current of 18. Assume that one relative-current unit corresponds to 0.10 mA.

1. Calculate the average current over one hour.
2. Estimate the ideal battery life in hours and in years.
3. Explain why the real service life would be shorter.
4. Suggest one change to the sensing or communication strategy that would extend autonomy.

Questions

1. Describe the information chain of an IoT system.
2. Why is edge processing useful for a battery-powered sensor?
3. Compare Wi-Fi, BLE and LoRaWAN for one application.
4. Why are software updates a safety and security requirement?
5. Give two methods for increasing sensor autonomy.

Engineering Challenge

Design a connected sensor for monitoring humidity in a school building. Select the sensing period, communication method and power source. Define one alarm rule, one local fallback function and two measures that protect user data.

Speaking Task

Should every household appliance be connected? Present one useful service, one environmental cost and one cybersecurity risk.

Sources

NIST – Internet of Things: online resource

LoRa Alliance: online resource

EC18 – Drones

How do autonomous aerial systems remain stable and safe?

Learning Objectives

- Explain how thrust, attitude and position are controlled in a multirotor drone.
- Identify the sensors used for stabilisation and navigation.
- Discuss endurance, payload, regulation and operational safety.

Engineering News

Drones are increasingly used for inspection, mapping, emergency response and precision agriculture. Their usefulness depends on more than flight performance: safe autonomy requires reliable sensing, energy management, communication and clearly defined procedures for fault conditions.

Main Article

A multirotor drone flies by controlling the thrust produced by several propellers. When the total upward thrust equals the weight, the aircraft hovers. Increasing all rotor speeds causes a vertical acceleration. Changing the balance between rotors creates roll, pitch or yaw moments and allows the drone to move horizontally or rotate.

The flight controller closes several feedback loops. Gyroscopes measure angular velocity and accelerometers measure specific acceleration. Their signals are combined to estimate attitude. A barometer provides altitude information, while satellite navigation and cameras can estimate position and horizontal velocity. Sensor fusion is essential because every sensor has limitations: gyroscopes drift, accelerometers are disturbed by vibration and GNSS signals may be inaccurate or unavailable indoors.

A common controller compares the desired attitude with the estimated attitude and adjusts motor speeds many times per second. The inner loop stabilises angular motion, while slower outer loops control velocity and position. If tuning is poor, the aircraft may oscillate or react too slowly to a gust.

Endurance is strongly limited by battery mass. A larger battery stores more energy but also increases weight and the power required to hover. Payload has the same effect. Engineers therefore optimise the frame, propellers, motors and flight plan together. Wind, temperature and repeated acceleration also reduce real endurance compared with ideal calculations.

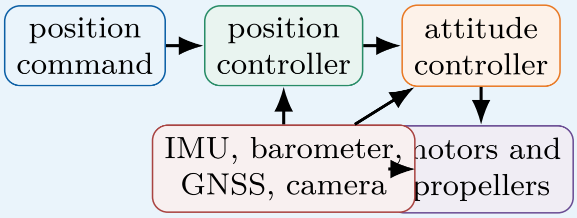
Autonomy introduces additional requirements. The drone must detect obstacles, maintain a safe separation from people and execute a predictable action if communication is lost. Depending on the mission, this may be hovering, landing or returning to a predefined point. Geofencing can prevent entry into restricted zones, but it does not replace pilot responsibility, maintenance or risk assessment.

Did You Know?

A quadrotor normally changes direction by tilting. Part of the total thrust then acts horizontally, so the motors must generate additional thrust to maintain altitude.

Engineering Focus

Nested control loops of a drone

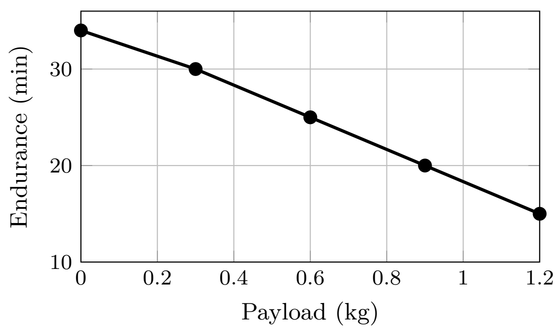


Fast attitude control stabilises the aircraft; slower position control generates the attitude command.

Key Figures

Hover condition	thrust = weight
Fast sensor	gyroscope
Altitude sensor	barometer
Main energy store	battery
Typical fault action	return or land

Payload and endurance



Illustrative trend for one aircraft: endurance decreases non-linearly as payload increases.

Useful Vocabulary

thrust	poussée
hover	vol stationnaire
roll / pitch / yaw	roulis / tangage / lacet
flight controller	contrôleur de vol
gyroscope	gyroscope
sensor fusion	fusion de capteurs
payload	charge utile
gust	rafale
geofencing	géorepérage
return-to-home	retour automatique

Application Exercise

The endurance graph gives 25 min for a payload of 0.6 kg. A safety rule requires the drone to land with 20% of the initial flight time still available. The outward journey to an inspection area takes 6 min and the return journey takes the same time.

1. Calculate the maximum authorised flight time before landing.
2. Determine the time available for inspection over the target area.
3. Recalculate this inspection time for a payload of 0.9 kg.
4. State two reasons why the pilot should use a larger reserve in strong wind.

Questions

1. How does a quadrotor produce roll and pitch motion?
2. Why is sensor fusion necessary?
3. Distinguish the attitude loop from the position loop.
4. Explain the compromise created by a larger battery.
5. What should happen after a communication failure?

Engineering Challenge

Define a drone mission to inspect a bridge. Estimate the required payload and flight time, select the main sensors, propose a safe path and define the response to low battery, GNSS loss and communication loss.

Speaking Task

Should drones be authorised to deliver parcels in cities? Compare efficiency, noise, safety, privacy and infrastructure requirements.

Sources

European Union Aviation Safety Agency – Drones: online resource

NASA – Unmanned Aircraft Systems: online resource

EC19 – Biomimetics

How can engineering learn from living systems?

Learning Objectives

- Distinguish inspiration from direct imitation of a biological model.
- Explain how structure, surface and control principles can be transferred to engineering.
- Evaluate a biomimetic solution using measurable performance and life-cycle criteria.

Engineering News

Biomimetic research is influencing adhesives, lightweight structures, ventilation systems and robot locomotion. Nature provides useful strategies, but engineering success requires abstraction, modelling and testing rather than simply copying the appearance of an organism.

Main Article

Biomimetics studies biological functions and transfers useful principles to engineered systems. The process normally begins with a technical problem such as reducing drag, gripping a surface or building a lightweight structure. Engineers then search for organisms that solve a comparable functional problem under similar constraints.

The biological solution must be understood at the correct scale. A bird wing is not only a curved surface; its performance depends on flexible feathers, muscles, sensing and active control. Directly copying its shape may therefore fail. Engineers abstract the underlying principle, create a model and adapt it to available materials and manufacturing methods.

Surface engineering provides many examples. Gecko feet adhere through millions of microscopic hairs that create weak intermolecular forces over a large contact area. Artificial dry adhesives reproduce the hierarchical structure without using glue. Shark skin has rib-like textures that influence the flow close to the surface. Similar riblets can reduce friction drag in selected conditions, although their benefit depends on size, orientation and cleanliness.

Nature also inspires structures. Bone combines a dense outer layer with a porous interior, giving a high stiffness-to-mass ratio. Additive manufacturing can produce lattice structures that use the same principle. Honeycomb panels and branching networks similarly offer efficient ways to distribute loads or fluids.

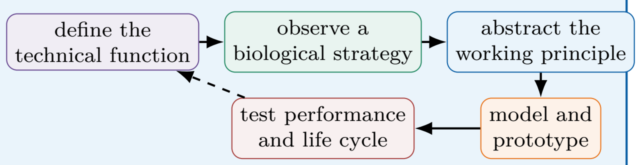
A biomimetic design is not automatically sustainable. Manufacturing a complex microstructure may consume large amounts of energy or use materials that are difficult to recycle. Engineers must compare the complete life cycle, durability and maintenance requirements with a conventional solution. Biomimetics is most valuable when the biological analogy leads to a measurable improvement and not only to an attractive product story.

Did You Know?

The hook-and-loop fastener was inspired by burrs that attached themselves to animal fur and clothing. The invention reproduced the attachment principle, not the complete plant structure.

Engineering Focus

From biological observation to engineering validation

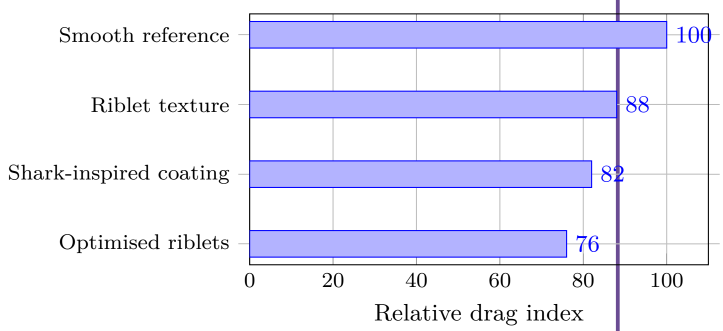


Iteration is essential because biological materials and manufacturing constraints are very different.

Key Figures

Primary objective	functional transfer
Common scale	microstructure
Typical benefit	low mass or drag
Necessary step	abstraction
Final proof	measured validation

Illustrative drag comparison



Illustrative values showing that geometry must be optimised for the operating flow.

Useful Vocabulary

biomimetics	biomimétisme
living system	système vivant
working principle	principe de fonctionnement
hierarchical structure	structure hiérarchique
riblet	micro-rainure
adhesion	adhérence
porous	poreux
lattice structure	structure en treillis
stiffness-to-mass ratio	rapport rigidité/masse
life cycle	cycle de vie

Questions

1. Why is direct imitation often insufficient?
2. Explain how gecko feet inspire dry adhesives.
3. What engineering benefit can be obtained from bone-like structures?
4. Why must operating scale be considered for riblets?
5. How can a biomimetic product have a poor environmental performance?

Engineering Challenge

Choose a biological strategy that could improve a small wind turbine, robot gripper or protective helmet. Identify the function, abstract the principle, propose a manufacturable concept and define two experiments that compare it with a conventional design.

Speaking Task

Is nature always the best engineer? Discuss optimisation, evolution, materials, manufacturing and the need for measurable evidence.

Sources

AskNature – Biomimicry examples: online resource
Biomimicry Institute: online resource

EC20 – Circular Economy

How can products remain useful for longer?

Learning Objectives

- Compare linear and circular product life cycles.
- Explain design for durability, repair, disassembly and remanufacturing.
- Use life-cycle reasoning to avoid shifting environmental impacts.

Engineering News

Manufacturers are developing repairable products, component take-back schemes and remanufacturing lines. Circularity is not achieved by recycling alone: the highest value is often preserved when a product, module or component remains in use with the least possible transformation.

Main Article

The traditional linear model follows a simple sequence: extract resources, manufacture a product, use it and discard it. This model loses materials, energy and technical value at the end of each life. A circular economy aims to keep products and materials useful for longer through maintenance, reuse, repair, refurbishment, remanufacturing and, finally, recycling.

Engineering decisions made during design strongly influence these options. A durable product needs components that resist expected loads and environments. A repairable product needs accessible fasteners, diagnostic information and available spare parts. Modular architecture allows a failed or obsolete function to be replaced without discarding the complete system.

Disassembly must also be considered. Permanent adhesives and mixed materials may simplify production but make separation difficult. Screws, clips and clearly identified polymers can improve recovery, although they may add cost, mass or assembly time. Engineers therefore balance circularity with safety, performance and industrial constraints.

Remanufacturing restores a used product to a defined level of performance. Components are inspected, cleaned, repaired or replaced and then tested. This can preserve more value than material recycling, but it requires traceability and a predictable return flow. Digital product passports can record materials, maintenance and previous use, helping operators choose the correct recovery route.

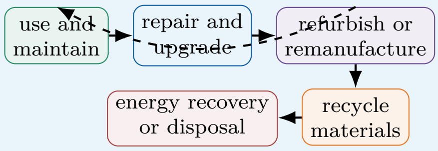
Life-cycle assessment prevents misleading solutions. A heavier product may last longer but require more material and transport energy. A replaceable battery can extend product life, yet the sealing system must still meet safety requirements. Engineers compare resource use, energy, emissions and service life across the complete system boundary. The objective is not simply to close a loop, but to reduce total environmental impact while providing the required service.

Did You Know?

Recycling is important, but it often destroys much of the manufacturing value already invested in a component. Repair and reuse usually preserve more value when they are technically possible.

Engineering Focus

Circular design hierarchy

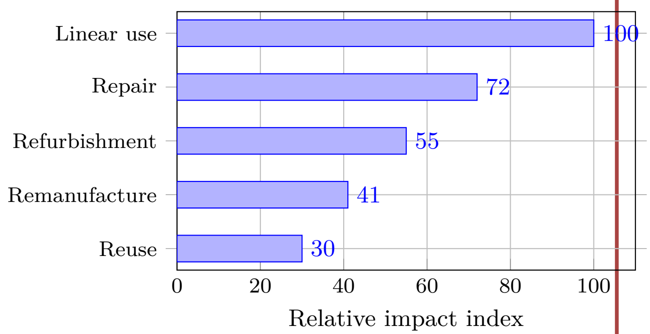


Short loops usually preserve more product value and require less transformation.

Key Figures

Highest-value strategy	continued use
Repair enabler	accessibility
Upgrade enabler	modularity
Recovery enabler	disassembly
Decision tool	life-cycle assessment

Relative life-cycle impact



Illustrative comparison for the same delivered service; real results depend on transport, energy mix, product lifetime and replacement parts.

Useful Vocabulary

linear economy	économie linéaire
durability	durabilité
repairability	réparabilité
disassembly	démontage
spare part	pièce détachée
refurbishment	reconditionnement
remanufacturing	remise à neuf industrielle
traceability	traçabilité
product passport	passeport produit
life-cycle assessment	analyse du cycle de vie

Questions

1. Why does repair usually preserve more value than recycling?
2. Give three design choices that improve repairability.
3. What is the difference between refurbishment and remanufacturing?
4. Why can permanent adhesive create an end-of-life problem?
5. How does life-cycle assessment prevent impact shifting?

Engineering Challenge

Redesign a cordless drill, laptop or coffee machine for a longer useful life. Define a modular architecture, select reversible joining methods, identify replaceable components and propose a collection and remanufacturing process. State one trade-off created by your design.

Speaking Task

Should manufacturers be legally required to provide spare parts and repair information? Discuss cost, safety, innovation, ownership and environmental impact.

Sources

European Commission – Circular Economy Action Plan: online resource
Ellen MacArthur Foundation – Circular Economy: online resource

EC21 – Digital Twin

Creating virtual replicas of physical systems

Learning Objectives

- Explain how measurements update a digital model.
- Distinguish simulation, monitoring and prediction.
- Identify the limits of a digital twin.

Engineering News

Digital twins are increasingly used for aircraft engines, factories, bridges and energy systems. By combining sensor data with physical or statistical models, engineers can estimate internal states, predict degradation and test operating strategies without disturbing the real asset.

Main Article

A digital twin is a virtual representation of a physical system that is regularly updated with real measurements. Unlike a static CAD model, it evolves with the asset. Sensors measure quantities such as temperature, vibration, force, pressure or electrical current. These data are filtered and compared with a mathematical model that describes how the system should behave.

The model may be physics-based, data-driven or hybrid. A physics-based model uses conservation laws, material properties and geometry. A data-driven model learns relationships from historical data. A hybrid twin combines both approaches: physics constrains the solution while machine learning corrects unknown effects or accelerates computation.

Engineers use twins for three main purposes. First, monitoring compares measured and expected behaviour. A growing difference can reveal sensor drift, wear or an incorrect model. Second, prediction estimates future states, for example remaining bearing life or battery temperature during a demanding mission. Third, optimisation allows engineers to test control settings, maintenance dates or production schedules before applying them to the real system.

A useful twin depends on data quality. Measurements must be time-stamped, calibrated and transmitted reliably. The model must also be validated over the full operating range. A twin that performs well at nominal speed may fail during start-up, overload or extreme weather. Uncertainty must therefore be reported rather than hidden.

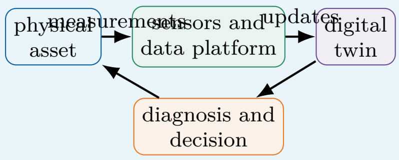
Digital twins do not replace physical tests. They help engineers decide which tests are most useful and interpret measurements more effectively. Their value comes from the continuous loop between the real asset, the data platform and the model.

Did You Know?

A model becomes a true digital twin only when information flows repeatedly between the physical asset and its virtual representation. Without this update loop, it is simply a simulation model.

Engineering Focus

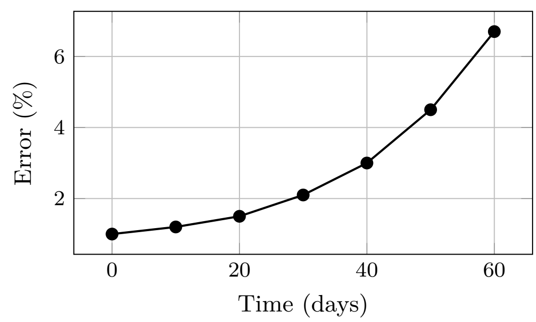
Information loop



Key Figures

Essential input	calibrated data
Core operation	model update
Typical output	health estimate
Main risk	model drift
Validation method	test comparison

Prediction error over time



A rising residual may indicate wear, a changing environment or an outdated model.

Useful Vocabulary

digital twin	jumeau numérique
asset	équipement réel
state estimation	estimation d'état
residual	résidu / écart
calibration	étalonnage
model drift	dérive du modèle
remaining useful life	durée de vie restante
uncertainty	incertitude
data-driven	fondé sur les données
validation	validation

Questions

1. How is a digital twin different from a CAD model?
2. What is the role of sensors in the update loop?
3. Compare physics-based and data-driven models.

4. Why must a twin be validated outside nominal operation?
5. What can cause the prediction error to increase?

Engineering Challenge

Design a digital twin for a school ventilation unit. Select four sensors, define the model outputs, propose one fault indicator and explain how the twin could reduce energy use without lowering air quality.

Speaking Task

Should safety-critical decisions ever be made automatically by a digital twin? Discuss reliability, responsibility, human supervision and uncertainty.

Sources

NIST – Digital Twins: online resource
NASA – Digital Twin: online resource

EC22 – Industry 4.0

Connected manufacturing

Learning Objectives

- Describe the architecture of a connected production system.
- Explain how data can improve availability and quality.
- Identify cybersecurity and human factors.

Engineering News

Factories are connecting machines, robots, inspection stations and planning software. The objective is not simply to automate more tasks, but to create a production system that can measure its own performance, adapt to variation and support faster decisions.

Main Article

Industry 4.0 describes the integration of connected machines, digital models, automation and data analysis in manufacturing. At machine level, sensors measure cycle time, motor current, vibration, position and product quality. Controllers operate the process in real time, while industrial networks send selected information to supervisory systems.

A connected factory is organised in layers. The physical layer contains machines and products. The control layer contains PLCs, drives and robot controllers. Manufacturing execution systems coordinate orders, traceability and work instructions. At the top, enterprise software manages resources, suppliers and customer demand. Data must move between these layers without compromising real-time operation.

Condition monitoring can detect a damaged bearing before failure. Vision systems can inspect every part instead of a small sample. Production data can reveal that a defect appears only with one tool, one material batch or one temperature range. These applications reduce unplanned downtime and waste, but only when engineers define useful indicators rather than collecting data without purpose.

Overall equipment effectiveness, or OEE, combines availability, performance and quality. A machine with high speed but frequent stops may have poor OEE. Connected monitoring helps teams identify the main loss and verify whether an improvement is effective.

Industry 4.0 also creates risks. A cyberattack can interrupt production or modify process parameters. Old machines may use insecure protocols. Operators may receive too many alarms or distrust opaque algorithms. Successful projects therefore combine secure architecture, clear interfaces, training and human expertise. Technology should support operators, not remove their understanding of the process.

Did You Know?

More data do not automatically produce better decisions. A useful industrial indicator must be linked to a physical cause, a decision and a person responsible for acting on it.

Engineering Focus

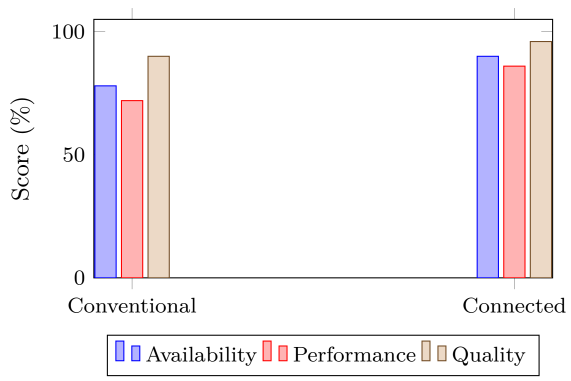
Connected production architecture

planning and enterprise
↕
MES and traceability
↕
PLCs and control
↕
machines and sensors

Key Figures

Availability loss	downtime
Performance loss	slow cycles
Quality loss	rejected parts
Typical local controller	PLC
Critical requirement	secure network

OEE components



Useful Vocabulary

shop floor	atelier de production
PLC	automate programmable
downtime	temps d'arrêt
traceability	traçabilité
work instruction	instruction de travail
condition monitoring	surveillance d'état
quality control	contrôle qualité
throughput	cadence / débit
legacy machine	machine ancienne
cybersecurity	cybersécurité

Questions

1. What are the main layers of a connected factory?
2. How can condition monitoring reduce downtime?
3. What three factors form OEE?

4. Why is collecting all possible data ineffective?
5. Name two human or cybersecurity risks.

Engineering Challenge

A packaging line suffers short, frequent stops. Propose a measurement plan, define three indicators, identify the required network connections and describe how operators would use the dashboard during a shift.

Sources

International Society of Automation: online resource
NIST – Smart Manufacturing: online resource

Speaking Task

Will connected factories create better industrial jobs or reduce human autonomy? Discuss safety, skills, productivity, surveillance and responsibility.

EC23 – Autonomous Vehicles

Engineering safe autonomous mobility

Learning Objectives

- Describe perception, localisation, planning and control.
- Explain why sensor redundancy is necessary.
- Analyse stopping distance and operational limits.

Engineering News

Autonomous driving systems are being tested in passenger cars, shuttles, mines, ports and warehouses. Progress is rapid in controlled environments, but safe operation on public roads remains difficult because weather, human behaviour and unusual situations are highly variable.

Main Article

An autonomous vehicle must perceive its environment, estimate its position, predict how other road users may move, plan a safe path and control steering, braking and propulsion. These functions operate continuously and must meet strict timing and safety requirements.

Perception combines cameras, radar, lidar and ultrasonic sensors. Cameras provide colour and detailed shapes but can be affected by darkness, glare or fog. Radar measures distance and relative speed and works well in poor visibility. Lidar creates accurate three-dimensional point clouds but may be costly and sensitive to rain or contamination. Sensor fusion combines their strengths and detects inconsistencies.

Localisation uses satellite positioning, inertial sensors, wheel odometry and maps. No single source is sufficient everywhere: GNSS signals can be blocked in tunnels, while wheel slip creates odometry errors. Redundant estimates allow the system to remain safe when one measurement becomes unreliable.

The planner selects a manoeuvre and generates a trajectory that respects road geometry, traffic rules, vehicle dynamics and comfort. The controller then follows this trajectory. Braking distance increases approximately with the square of speed because kinetic energy is proportional to v^2. Wet roads, tyre condition and reaction delays increase the total stopping distance.

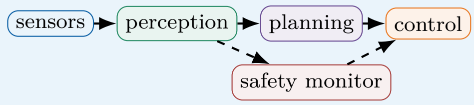
A safe system also needs an operational design domain: the conditions in which automation is allowed. These conditions may specify road type, speed, weather and map quality. When the vehicle reaches the boundary of this domain, it must request human control or perform a minimal-risk manoeuvre. Safety therefore depends not only on intelligence, but also on clear limits and predictable fallback behaviour.

Did You Know?

A vehicle may correctly detect an object yet still choose an unsafe action. Verification must therefore cover the complete chain from sensing to decision and control.

Engineering Focus

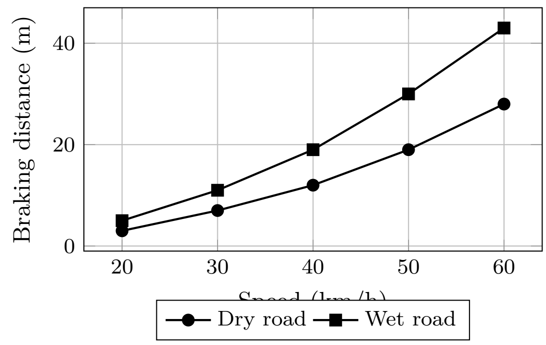
Autonomous driving pipeline



Key Figures

Camera strength	detailed images
Radar strength	relative speed
Lidar output	3D point cloud
Critical principle	redundancy
Safe fallback	minimal-risk manoeuvre

Illustrative braking distance



Useful Vocabulary

sensor fusion	fusion de capteurs
point cloud	nuage de points
localisation	localisation
trajectory	trajectoire
fallback	solution de repli
operational design domain	domaine opérationnel
wheel slip	glissement des roues
blind spot	angle mort
braking distance	distance de freinage
redundancy	redondance

Questions

1. What functions form the autonomous driving chain?
2. Why are several sensor technologies combined?
3. How can localisation fail in a tunnel?
4. Why does braking distance rise rapidly with speed?
5. What is an operational design domain?

Engineering Challenge

Design an autonomous shuttle for a closed university campus. Define its sensors, maximum speed, operational design domain, emergency strategy and verification tests for pedestrians, rain and GNSS loss.

Speaking Task

Who should be responsible after an autonomous vehicle accident: the passenger, manufacturer, software provider or road operator? Defend a position.

Sources

SAE International – Automated Driving: online resource

NHTSA – Automated Vehicles: online resource

EC24 – Smart Grids

Managing tomorrow's electricity networks

Learning Objectives

- Explain how generation and demand must remain balanced.
- Describe the role of storage, forecasting and demand response.
- Identify resilience and cybersecurity constraints.

Engineering News

Electricity networks are changing from one-way systems dominated by large power stations into interactive networks containing solar roofs, wind farms, batteries, electric vehicles and flexible loads. Digital control is becoming essential to coordinate these distributed resources.

Main Article

An electricity grid must balance production and consumption at every moment. If generation is lower than demand, frequency falls; if it is higher, frequency rises. Traditional grids adjusted a limited number of controllable power stations. Smart grids coordinate many smaller and more variable devices.

Smart meters measure energy flows at short intervals. Forecasting software estimates demand and renewable generation. Automated substations can change network configuration after a fault, while batteries and flexible loads respond rapidly to imbalances. Demand response shifts selected uses in time: for example, water heating or electric-vehicle charging may be delayed when the grid is heavily loaded.

Distributed generation creates bidirectional power flows. A neighbourhood may consume electricity in the evening but export solar power at midday. Voltage can rise near photovoltaic installations, and protection systems must distinguish normal reverse flow from a dangerous fault. Engineers therefore combine power electronics, communication systems and network models.

Storage provides several services. Batteries can smooth short fluctuations, support frequency and reduce peak demand. Pumped hydro stores larger amounts of energy for longer periods. Electric vehicles may eventually provide controlled charging or energy back to the grid, but battery degradation and user mobility must be respected.

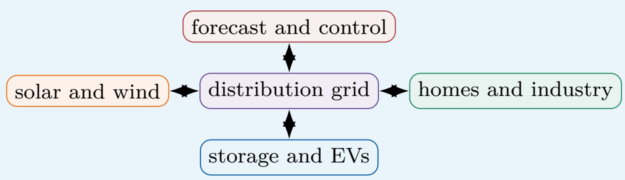
Digitalisation increases exposure to cyberattack and communication failure. Local controllers must retain safe behaviour when the central platform is unavailable. Resilience also requires physical redundancy, black-start capability and plans for extreme weather. A smart grid is not merely connected; it must remain stable, secure and understandable during abnormal conditions.

Did You Know?

Electricity itself travels through the network almost instantly, but energy planning may cover seconds, hours, seasons and years. Different storage technologies are needed for these different timescales.

Engineering Focus

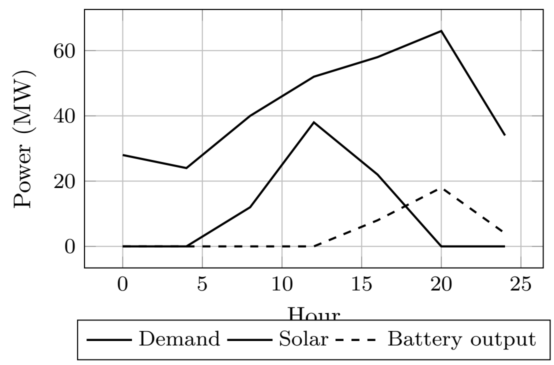
Bidirectional smart grid



Key Figures

Instant requirement	power balance
Fast flexibility	batteries
Long storage	pumped hydro
Flexible demand	smart charging
Critical risk	cyberattack

Daily balancing example



Useful Vocabulary

smart meter	compteur communicant
demand response	effacement / flexibilité
distributed generation	production décentralisée
peak demand	pointe de consommation
frequency control	réglage de fréquence
bidirectional flow	flux bidirectionnel
substation	poste électrique
black start	redémarrage autonome
load shedding	délestage
grid resilience	résilience du réseau

Questions

1. Why must generation equal demand continuously?
2. How does demand response help the network?
3. What problem can midday solar export create?
4. Compare batteries and pumped hydro.

5. Why should local controllers work without the central platform?

Engineering Challenge

Plan the energy management of a school with rooftop solar panels, a 200 kWh battery and ten EV chargers. Define priorities for a normal day, a winter peak and a two-hour grid outage.

Speaking Task

Should electricity suppliers be allowed to delay domestic appliances automatically during peak demand? Discuss consent, comfort, price, privacy and grid stability.

Sources

International Energy Agency – Smart Grids: online resource

ENTSO-E: online resource

EC25 – Marine Renewable Energy

Power from oceans

Learning Objectives

- Compare tidal, wave and offshore energy technologies.
- Explain the mechanical and environmental design constraints.
- Analyse energy yield and maintenance trade-offs.

Engineering News

Oceans contain large renewable energy resources. Offshore wind is already deployed at commercial scale, while tidal-stream and wave-energy devices continue to develop. Marine systems benefit from strong resources but face difficult installation and maintenance conditions.

Main Article

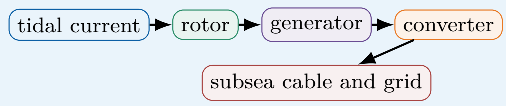
Marine renewable energy includes offshore wind, tidal barrages, tidal-stream turbines and wave-energy converters. Each technology extracts energy from a different physical phenomenon and therefore needs a different structure, control system and maintenance strategy.
Tidal-stream turbines operate like underwater wind turbines. Water density is much greater than air density, so a relatively slow current can produce significant force. The available power is proportional to fluid density, swept area and the cube of flow speed. Tides are highly predictable, but suitable sites are limited and the equipment must resist strong cyclic loads.
Wave-energy converters use the relative motion between waves and a floating or fixed structure. Some devices use oscillating water columns, while others use hinged bodies or hydraulic power take-off systems. Wave direction, period and height change continuously, making efficient control difficult. Storm survival is often more demanding than normal power production.
The marine environment accelerates corrosion and biological growth. Salt water attacks materials, while barnacles and algae change surface properties and increase drag. Seals, cables and bearings must operate for long periods with limited access. Retrieving a device for repair may require specialised ships and suitable weather, so reliability strongly affects the cost of energy.
Engineers must also consider navigation, fishing, underwater noise and marine ecosystems. Environmental monitoring is required before and after installation. Grid connection can be expensive because subsea cables and offshore substations are needed. The best design is therefore not simply the device with the highest peak power, but the system that delivers reliable energy over its complete life cycle.

Did You Know?

Because tidal motion is driven by astronomical cycles, its timing can be predicted years in advance. This makes tidal energy variable but highly predictable.

Engineering Focus

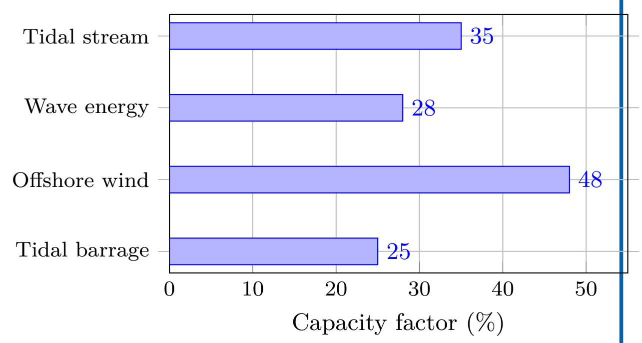
Tidal-stream conversion chain



Key Figures

Predictable resource	tides
Main material threat	corrosion
Common surface issue	biofouling
Major cost driver	offshore maintenance
Grid connection	subsea cable

Illustrative capacity factors



Values are illustrative; real performance depends strongly on site and technology maturity.

Useful Vocabulary

tidal stream	courant de marée
wave period	période de vague
power take-off	système de conversion d'énergie
cyclic load	charge cyclique
corrosion	corrosion
biofouling	encrassement biologique
subsea cable	câble sous-marin
capacity factor	facteur de charge
mooring	amarrage
storm survival	tenue en tempête

Questions

1. Why can slow seawater flows produce large forces?
2. What makes wave-energy control difficult?
3. Why does maintenance dominate marine energy costs?
4. Give two environmental impacts that must be monitored.
5. Why is peak power insufficient for comparing devices?

Engineering Challenge

Choose a tidal-stream site and propose a 1 MW device. Estimate the required rotor area for an assumed current speed, select corrosion protection, define a maintenance strategy and identify two ecological monitoring measures.

Speaking Task

Should marine renewable projects be prioritised in protected coastal areas when they reduce carbon emissions? Discuss biodiversity, energy security, local jobs and precaution.

Sources

International Energy Agency – Ocean Power: online resource

European Marine Energy Centre: online resource

EC26 – Biomedical Imaging

Seeing inside the human body

Learning Objectives

- Explain the main physical and engineering principles.
- Interpret measurements, models and performance trade-offs.
- Identify safety, environmental and validation constraints.

Engineering News

Biomedical imaging allows clinicians to observe anatomy and function without surgery. Engineers combine wave physics, detectors, signal processing and reconstruction algorithms to produce useful images while controlling acquisition time, cost and patient risk.

Main Article

Biomedical imaging converts physical interactions inside the body into maps that clinicians can interpret. Different technologies measure different properties. X-ray radiography and computed tomography use the attenuation of ionising radiation. Ultrasound sends mechanical waves and analyses their echoes. Magnetic resonance imaging, or MRI, measures signals produced by hydrogen nuclei in a strong magnetic field. Nuclear medicine detects radiation emitted by a tracer introduced into the body.

Every imaging chain contains four stages: excitation, interaction, detection and reconstruction. The detector rarely produces a ready-to-use picture. It records many noisy signals from different positions or times. A reconstruction algorithm estimates the spatial distribution that best explains these measurements. Increasing spatial resolution usually requires more data, more acquisition time or a stronger signal.

Image quality is described by resolution, contrast, signal-to-noise ratio and artefacts. Resolution determines the smallest visible detail. Contrast separates tissues with similar properties. Noise creates random variations, while artefacts create structured errors caused by motion, metal objects or incomplete measurements. Engineers must choose acquisition parameters that balance these quantities.

Safety is equally important. X-ray dose must be kept as low as reasonably achievable. MRI avoids ionising radiation but requires strict control of magnetic objects, radio-frequency heating and acoustic noise. Ultrasound is portable and relatively inexpensive, yet image quality depends strongly on the operator and on acoustic access to the organ.

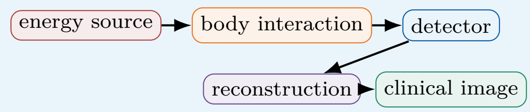
Artificial intelligence can assist segmentation, reconstruction and anomaly detection, but it must be validated on diverse patients and scanners. A convincing image is not necessarily an accurate one. Biomedical engineers therefore work closely with physicists, radiologists, clinicians and regulators.

Did You Know?

A high-performance prototype is not sufficient: engineers must prove repeatability, safety and useful performance under realistic operating conditions.

Engineering Focus

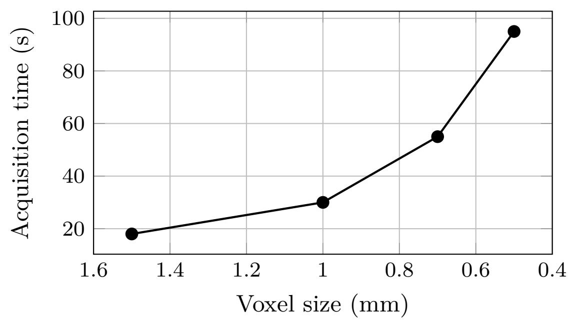
From measurement to diagnosis



Key Figures

X-ray / CT	attenuation
Ultrasound	echo time
MRI	magnetic resonance
Main trade-off	quality vs risk
Critical step	validation

Resolution and acquisition time



Useful Vocabulary

attenuation	atténuation
echo	écho
voxel	voxel
spatial resolution	résolution spatiale
contrast	contraste
noise	bruit
artefact	artefact
reconstruction	reconstruction
radiation dose	dose de rayonnement
tracer	traceur

Questions

1. Why do different imaging technologies reveal different tissues?
2. What is the role of reconstruction algorithms?
3. Explain the difference between noise and an artefact.
4. Why is higher resolution not always the best choice?
5. Give one safety constraint for CT and one for MRI.

Engineering Challenge

Design an imaging protocol for a moving organ. Choose a technology, define the required resolution and acquisition time, identify one major artefact and propose a method to reduce it.

Speaking Task

Should AI-generated medical images be accepted when they improve visibility but modify the original measurements? Discuss accuracy, responsibility and patient safety.

Sources

World Health Organization – Medical imaging: online resource

International Atomic Energy Agency – Radiation protection: online resource

EC27 – Nanotechnology

Engineering at the nanoscale

Learning Objectives

- Explain the main physical and engineering principles.
- Interpret measurements, models and performance trade-offs.
- Identify safety, environmental and validation constraints.

Engineering News

At dimensions below about one hundred nanometres, surfaces, interfaces and quantum effects can dominate material behaviour. Nanotechnology uses these effects in electronics, coatings, sensors, medicine and energy systems.

Main Article

A nanometre is one billionth of a metre. At this scale, a material can behave differently from the same substance in bulk form. Two reasons are especially important. First, the surface-area-to-volume ratio becomes very large, so surface reactions and interfaces dominate. Second, electrons may be confined to very small regions, creating size-dependent optical and electrical properties.

Engineers use two broad manufacturing strategies. Top-down methods remove material from a larger structure by lithography, etching or machining. Bottom-up methods assemble structures from atoms, molecules or nanoparticles through chemical synthesis and self-organisation. Top-down fabrication offers precise patterns, while bottom-up processes can produce very small features over large areas.

Nanomaterials include particles, tubes, wires, thin films and porous structures. Carbon nanotubes combine low density with high stiffness and electrical conductivity. Quantum dots emit colours that depend on particle size. Nanostructured catalysts expose many active sites, reducing the quantity of expensive material required. Thin coatings can modify friction, corrosion, wettability or optical reflection without changing the bulk component.

Measurement is difficult because ordinary optical microscopes cannot resolve features much smaller than visible-light wavelengths. Electron microscopy, atomic-force microscopy and spectroscopy are therefore essential. Manufacturing control must also prevent agglomeration, contamination and variation in particle size.

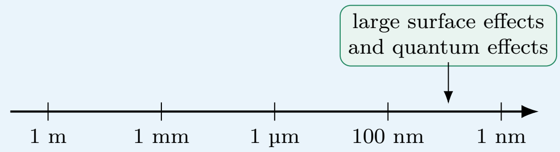
Benefits must be assessed together with risks. Nanoparticles can enter organisms and ecosystems in ways that larger particles cannot. Toxicity depends on size, shape, coating and chemical composition, not only on mass. Engineers must therefore design containment, exposure monitoring, recycling and end-of-life procedures. Responsible nanotechnology requires lifecycle thinking as well as impressive laboratory performance.

Did You Know?

A high-performance prototype is not sufficient: engineers must prove repeatability, safety and useful performance under realistic operating conditions.

Engineering Focus

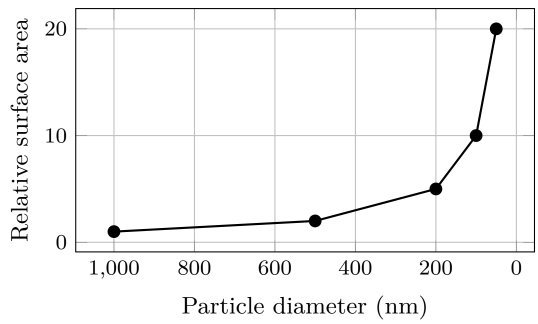
Scale changes behaviour



Key Figures

1 nanometre	10^{-9} m
Typical nanoparticle	1–100 nm
Top-down	material removal
Bottom-up	self-assembly
Main concern	exposure

Surface area increases as size decreases



Useful Vocabulary

nanoscale	échelle nanométrique
nanoparticle	nanoparticule
thin film	couche mince
lithography	lithographie
etching	gravure
self-assembly	auto-assemblage
quantum dot	boîte quantique
agglomeration	agglomération
coating	revêtement
exposure	exposition

Questions

1. Why do surfaces become more important at the nanoscale?
2. Compare top-down and bottom-up manufacturing.
3. Why are electron microscopes often required?
4. Give two useful properties of nanostructured materials.

5. Why is particle mass insufficient for evaluating risk?

Engineering Challenge

Select a conventional product and propose a nanostructured coating that improves one function. Define the manufacturing route, verification method, expected benefit and exposure-control measures.

Sources

European Commission – Nanomaterials: online resource
National Nanotechnology Initiative: online resource

Speaking Task

Do the potential benefits of nanoparticles justify their use before all long-term environmental effects are known?

EC28 – Quantum Technologies

Using superposition, entanglement and precise measurement

Learning Objectives

- Explain the main physical and engineering principles.
- Interpret measurements, models and performance trade-offs.
- Identify safety, environmental and validation constraints.

Engineering News

Quantum technologies are moving from laboratory demonstrations to specialised sensors, secure communication links and prototype computers. Their advantage comes from controlling individual quantum states rather than simply making conventional devices smaller.

Main Article

Quantum technologies exploit physical effects that are negligible in ordinary engineering systems. A quantum bit, or qubit, can be prepared in a superposition of two basis states. Several qubits may also become entangled, meaning that their measurement results are correlated more strongly than classical systems allow. These properties enable new methods of computation, sensing and communication.

A quantum computer does not test every answer at once. It applies carefully designed operations to amplitudes and uses interference to increase the probability of useful results. Measurement then produces a classical bit string. Quantum algorithms can accelerate certain tasks such as factoring, molecular simulation or specific optimisation subroutines, but they do not make every calculation faster.

The main engineering difficulty is decoherence. Qubits interact unintentionally with their environment and lose the fragile information stored in their state. Platforms based on superconducting circuits, trapped ions, neutral atoms and photons therefore require precise control, isolation and calibration. Some need cryogenic temperatures; others need ultra-high vacuum and stable lasers.

Quantum error correction distributes one logical qubit across many physical qubits. It can detect and correct errors without directly reading the encoded information, but the hardware overhead is large. Current machines are therefore described as noisy and intermediate-scale. They are valuable research tools but not yet universal fault-tolerant computers.

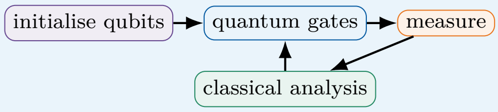
Quantum sensing is closer to deployment in some fields. Atomic clocks, magnetometers and gravimeters use quantum transitions to measure time, magnetic field or acceleration with extreme sensitivity. Quantum communication can reveal interception because measurement disturbs an unknown quantum state. Engineering success will depend on systems integration, standards, realistic benchmarking and secure links between quantum and classical equipment.

Did You Know?

A high-performance prototype is not sufficient: engineers must prove repeatability, safety and useful performance under realistic operating conditions.

Engineering Focus

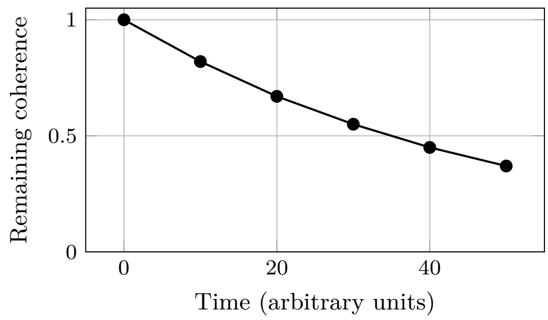
Quantum computing chain



Key Figures

Information unit	qubit
Main resource	coherence
Common enemy	noise
Output	probability distribution
Near-term strength	sensing

Illustrative coherence decay



Useful Vocabulary

qubit	qubit
superposition	superposition
entanglement	intrication
interference	interférence
measurement	mesure
decoherence	décohérence
quantum gate	porte quantique
error correction	correction d'erreurs
cryogenic	cryogénique
fault-tolerant	tolérant aux fautes

Questions

1. How does a qubit differ from a classical bit?
2. Why does measurement limit the information obtained from a quantum state?
3. What causes decoherence?
4. Why are many physical qubits needed for one logical qubit?
5. Name one application of quantum sensing.

Engineering Challenge

Propose an experimental quantum sensor for detecting small magnetic-field variations. Describe the quantum element, control system, environmental isolation, calibration and classical signal processing.

Speaking Task

Should governments prioritise quantum computing, quantum sensing or quantum-secure communication? Defend one investment strategy.

Sources

NIST – Quantum information science: online resource
European Quantum Flagship: online resource

EC29 – Sustainable Buildings

Designing low-carbon and comfortable spaces

Learning Objectives

- Explain the main physical and engineering principles.
- Interpret measurements, models and performance trade-offs.
- Identify safety, environmental and validation constraints.

Engineering News

Buildings consume energy during construction and throughout decades of operation. Sustainable design therefore combines passive architecture, efficient equipment, low-carbon materials, renewable energy and monitoring of real performance.

Main Article

A sustainable building must provide comfort, safety and useful space while reducing environmental impacts over its entire lifecycle. Operational energy is used for heating, cooling, ventilation, lighting and equipment. Embodied carbon is associated with extracting materials, manufacturing products, transport, construction, maintenance and demolition. As grids become cleaner and insulation improves, embodied impacts represent a larger share of the total.

The first design strategy is to reduce demand. Building orientation, window area, shading and thermal mass influence solar gains. Insulation and airtightness reduce heat transfer, while controlled ventilation maintains air quality. External shading can block summer sun without preventing winter daylight. These passive decisions are usually more robust than adding complex equipment later.

Heating and cooling systems must then match the reduced demand. Heat pumps can transfer several units of heat for one unit of electricity, but their performance depends on outdoor temperature and supply temperature. Efficient fans, pumps and controls reduce auxiliary consumption. Sensors can adjust operation according to occupancy, carbon-dioxide concentration and weather forecasts.

Material selection requires lifecycle assessment. Concrete offers durability and thermal mass but cement production emits carbon dioxide. Timber stores biogenic carbon and can reduce mass, yet sourcing, fire safety, moisture and end-of-life must be considered. Reuse of structural elements can avoid both waste and new production.

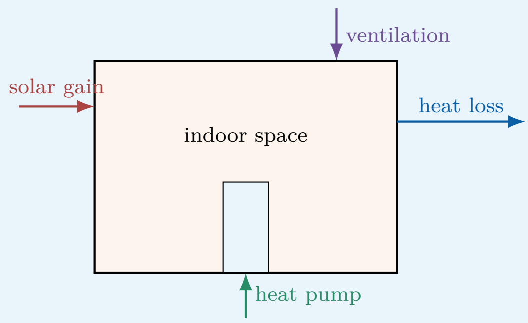
A design model is only a prediction. The performance gap between calculated and measured consumption may result from construction defects, incorrect controls or occupant behaviour. Commissioning, metering and feedback are therefore part of sustainable engineering. A successful building is not merely certified at handover; it continues to perform well during use.

Did You Know?

A high-performance prototype is not sufficient: engineers must prove repeatability, safety and useful performance under realistic operating conditions.

Engineering Focus

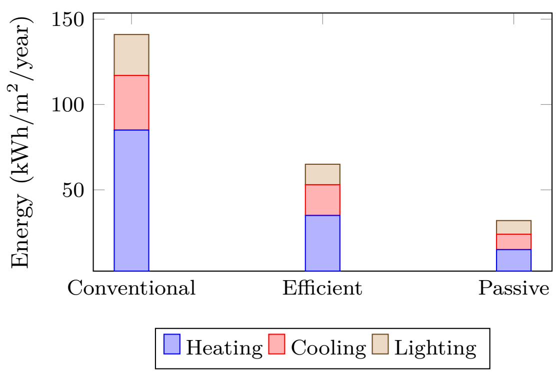
Heat and energy flows



Key Figures

First priority	reduce demand
Envelope metric	U-value
Air quality proxy	CO_2
Operational check	metering
Material tool	lifecycle assessment

Illustrative annual energy demand



Useful Vocabulary

embodied carbon	carbone incorporé
operational energy	énergie d'exploitation
airtightness	étanchéité à l'air
thermal bridge	pont thermique
shading	protection solaire
heat pump	pompe à chaleur
commissioning	mise au point
metering	comptage
performance gap	écart de performance
lifecycle assessment	analyse du cycle de vie

Questions

1. What is the difference between operational and embodied carbon?
2. Why should demand reduction precede equipment selection?
3. How can external shading improve summer comfort?
4. Why can actual consumption differ from simulation?
5. Give two reasons for monitoring indoor carbon dioxide.

Engineering Challenge

Design a low-energy classroom for a temperate climate. Propose envelope targets, shading, ventilation, heating, sensors and a method for checking actual performance after one year.

Speaking Task

Should building regulations limit embodied carbon as strictly as operational energy?

Sources

International Energy Agency – Buildings: online resource

UN Environment Programme – Buildings and construction: online resource

EC30 – High-Speed Transportation

Moving faster with less energy

Learning Objectives

- Explain the main physical and engineering principles.
- Interpret measurements, models and performance trade-offs.
- Identify safety, environmental and validation constraints.

Engineering News

High-speed rail, magnetic-levitation systems and evacuated-tube concepts seek to reduce journey times between cities. At high speed, aerodynamic drag, infrastructure quality, safety and energy supply become dominant engineering constraints.

Main Article

High-speed transportation is a system problem rather than simply a faster vehicle. Track geometry, propulsion, signalling, stations, maintenance and emergency procedures must be designed together. A train travelling at 300 km/h covers more than 80 metres each second, so small errors can rapidly become critical.

Aerodynamic drag increases approximately with the square of speed, while the power required to overcome it increases approximately with the cube. Doubling speed can therefore require about eight times more aerodynamic power if shape and air density remain unchanged. Streamlined noses, smooth surfaces and protected underbodies reduce drag and pressure waves. Tunnel entry is particularly demanding because the vehicle compresses air in a confined space.

Conventional high-speed trains use steel wheels on rails. This provides low rolling resistance and mature switching technology. Magnetic levitation removes wheel-rail contact at speed, reducing mechanical wear, but it requires dedicated guideways and complex electromagnetic control. Low-pressure tubes could reduce aerodynamic resistance further, yet maintaining long evacuated infrastructure and safe evacuation routes is difficult.

Braking distance grows with the square of speed for a given deceleration. High-speed networks therefore use continuous signalling and automatic train protection rather than relying on lineside signals. Regenerative braking can return energy to the electrical network, provided another train or storage system can absorb it.

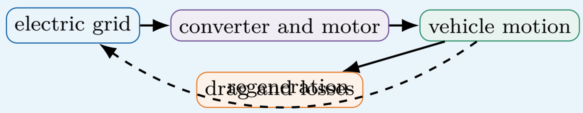
Speed alone does not determine usefulness. Door-to-door time, station access, capacity, reliability, construction carbon and ticket cost matter. High-speed rail performs best on busy corridors where large passenger flows justify dedicated infrastructure. Engineers must compare alternatives using energy per passenger-kilometre and lifecycle impact, not vehicle speed or peak power alone.

Did You Know?

A high-performance prototype is not sufficient: engineers must prove repeatability, safety and useful performance under realistic operating conditions.

Engineering Focus

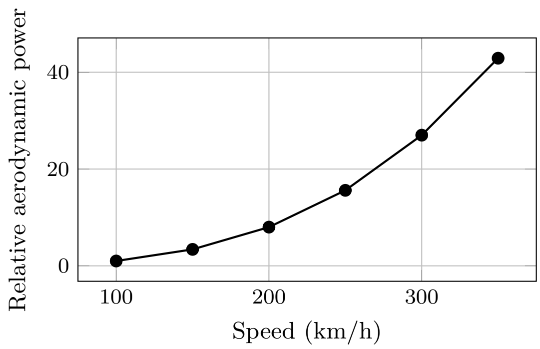
Energy conversion in high-speed rail



Key Figures

Drag force	$\propto v^2$
Aerodynamic power	$\propto v^3$
Braking distance	$\propto v^2$
Safety system	continuous signalling
Useful metric	passenger-km

Aerodynamic power versus speed



Useful Vocabulary

aerodynamic drag	traînée aérodynamique
rolling resistance	résistance au roulement
guideway	voie de guidage
magnetic levitation	sustentation magnétique
pressure wave	onde de pression
braking distance	distance de freinage
regenerative braking	freinage régénératif
signalling	signalisation
passenger-kilometre	passager-kilomètre
dedicated infrastructure	infrastructure dédiée

Questions

1. Why does aerodynamic power rise so quickly with speed?
2. What are the main advantages and limits of magnetic levitation?
3. Why are continuous signalling systems necessary?
4. When can regenerative braking return useful en-

ergy?

5. Why should door-to-door time be considered?

Engineering Challenge

Compare a 300 km/h railway with a magnetic-levitation line for a 600 km corridor. Estimate travel time, identify dominant losses, define safety requirements and propose a lifecycle comparison.

Speaking Task

Should public investment favour maximum speed or a denser network with lower speeds and more frequent services?

Sources

International Union of Railways – High-speed rail: online resource

International Energy Agency – Rail: online resource